2023

CALENDAR YEAR REPORT

WASTEWATER TREATMENT FACILITIES

City of Memphis

Division of Public Works

Prepared by Don Hudgins, P.E.

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2023 Calendar Year Report
Wastewater Treatment Facilities
City of Memphis

1.0 Maxson Wastewater Treatment Facility

Introduction-

Maxson Facility dewatering and Biorem system pipes burst during a cold snap due to rolling power black outs as MLGW continued to struggle meeting Memphis power demand due to TVA cutbacks. TDEC inspection was on 1-23-2023. On 1-26-2023, tremendous white foam in the aeration basins and coming over the weirs of the disinfection tank occurred, a condition that continued into February. TDEC emailed their CEI Report on 2-27-2023, including an NOV for QNC during November and December 2022.

In March, the Judge dismissed the HLSD lawsuit in summary judgement. The Judge ruled that the contract with the City ended on 9-22-2023 and the City still had an obligation to treat HLSD wastewater so that matter would proceed to trial to determine time and money. The draft AOC for Maxson was reviewed. The foam finally began dissipating along with improvement in plant processes associated with lower loading due to curtailed production at Solae. However, aeration basin DWPs continued pegging indicating the need for regular cleaning. The effluent strainer investigation was wrapped up with the idea that strained disinfected effluent could be put to non-potable use around the plant. On 3-27-2023, the Maxson Facility started using both sides for disinfection to handle occasional high flows (>85 mgd).

On 4-6-2023, a sinkhole took out Plant Road coming out of the park exposing a 24-inch water line running parallel and fiber optic and telephone lines running perpendicular to the road. The 96-inch Nonconnah Interceptor had failed at the road crossing. The pipe at the road crossing was an Armco Smooth-Flo product 100% lined with asphalt. Soon after the sink hole formed, remnants of the Armco Smooth-Flo product, including corrugated steel, began showing up on the coarse bar screens. The Influent %VSS began going down right before the cavity formed indicating sand intrusion. Based on the influent data, the failure started possibly as early as mid-March. All CBS were taken out of service after being hit with debris from sewer failure and the need to increase raw sewage pumping during rain events to keep from surcharging cavity was identified. The pipe likely gave way due to internal pressure during countless surcharged conditions over 50+ years.

Planning continued for the solar farm project looking at the possible generation of 8 MW to handle the power needs of the plant on sunny days. Changes needed to be made to the Emerson DCS to handle the new trickling filter equipment was identified but lack of coordination between the Emerson group employed to install the Emerson equipment in the new Admin Building and the Emerson group connecting installing controls and sensors in the new trickling filters and other new treatment units would prove problematic.

On 4-24-23, a test run taking Stiles biosolids to Maxson was performed by Denali. The width of the fingers in the fields may need to be increased to accommodate dump trailers. Keeping MLSS in the 4200-5200 mg/l range to promote BOD removal which has reduced foaming resulted in poor settling. Rocks from the floor of Primary Clarifier #2 clarifier floor which had been

removed from service were still getting into and clogging bio-pumps. Edge approved a tax abatement for Richard Oil Seed to improve and enlarge its 1351 Williams facility.

On 5-8-2023, a tree limb hung up in the CBS possibly from the cavity. Most of the sink hole itself appears to have shown up in the grit tank. Polymer trials were conducted on 5-10/11-23 with only Polydyne showing interest. The Maxson Facility received NOV QNCR for January-April permit excursions on 5-19-2023.

Solae experienced limited production since beginning of the month resulted in dissolved oxygen levels = 5 mg/l+ and DWP = 31. Xylem continued fusing pipe with the bypass scheduled to start in June. The trial with HLSD ended in May with judgement expected before the contract end date of 9-22-2023. Erika Collins emailed on 6-5-2023 that Solae experienced unplanned/unscheduled downtime and will have minimal/zero discharge for next several days due to an employee strike. Based on their SMRs, the work slowdown possibly started in early May. The month of June therefore would be a good month to determine loading baseline without Solae as demonstrated later in this report. On 6-13-2023, 10-12 loads of Stiles Facility biosolids arrived at the Maxson Facility on 6-13-2023 by Denali for surface disposal. On 6-15-2023, Cable Tel hit the 16-inch primary sludge wasting line and primary sludge wasting was stopped. A limited amount of sludge was released on a roadway within the main plant which was vactored and TDEC was notified on 6-15-2023. A slug discharge was reported by Van Guard Soap/Valley Products due to a leaking rail car on about 6-26-2023 and slowly began coating the primary clarifiers. Approximately 1010 lbs of muristic acid leaked during the night of 6-26-2023. Operations noted slimy samples on 6-27-2023 during 3-11 shift. Lab notified Process Dept on 6-28-2023. Greasy foam in primary clarifiers, aeriation basins, and disinfection basin was observed on 6-30-2023. A 4th blower could not be turned on 6-30-2023 due to surging. Scum began coating the clarifiers and breaking up and strong sulfur odor permeated the plant. At the same time, a heavy layer of grease continued to build pushing up into MH9 riser due to surcharging related to the sewer line collapse and eventually began accumulating in the crown of the pipe later growing to gigantic proportions. The Nonconnah Bypass was expected to start toward end of July.

Ten (10) power poles were knocked down on 7-5-2023 on Plant Road due to high wind, knocking out power forcing the Maxson Facility to switch to the south circuit. Maxson Facility began having trouble starting a second blower without surging. Before the storm, all 4 blowers could be started. Fouling of the FBD was suspected of restricting air flow as well as starter problems. TDEC was notified of the incident on 7-7-2023. The second blower was finally backup on 7-8-2024.

Influent of unusual character began entering the plant after 7-4-2023. Stringy material was observed by lab permeating samples on 7-7, 7-10, and 7-13. This could have been due to hotter temps causing the backed-up FOG to start dissolving and coming into the treatment plant. Sulfur loading was increasing at the plant based on testing results as well has higher H2S levels. Maxson had sudden drop in D.O. on 7-11-2023 due to a combination of heavy loading and FBD coating. The diffusers were sent to lab for evaluation. Meanwhile, the new D.O. probes began malfunctioning due to fouling. While the existing 3 TFs continued to run, TF1 arms began failing to corrosion of distribution arms and the system rigged to turn. Molybdenum levels in biosolids were observed creeping up based on monthly sampling.

Maxson Facility had plant-wide power outage at 7 pm on 7-25-2023 as result of electric power blip across the City. Power back up at 11 pm. An estimated 1.7 million gallons overflowed from the low manhole and TDEC was notified. Cleaning FBD with scrub brushes and soap was initiated to improve aeration.

In August, the giant grease plug grew upstream to MH9 as construction of the Nonconnah Bypass continued. The question is asked, why is grease a problem here and not at the upstream siphon at I-240? Installation of the CBS rake continued to be hindered by the need not to back up the flow on the Nonconnah Interceptor due to sinkhole. EDI evaluation of FBD showed the coating to be half polar and half non-polar O&G.

On 8-10-2023, high flow at 7am caused bypass area to go underwater due to influent raw sewage pumping. Only one RSP remained available (#2). Xylem had not reinstalled the 48-inch riser which allowed sewage to enter the excavation at a lower elevation possibly contributing to more loose rock and gravel and TDEC notified.

As an outgrowth of the effluent strainer evaluation, a big push was made for the Maxson Facility to give effluent water to TVA to get the TVA Combined Cycle plant off MLGW potable water and apparently TVA was receptive to the idea having hired Veolia as their engineer. TVA apparently had a change of heart because they originally turned down the project since it did not fit their mission statement to provide economical power. The Solar Project PER should be wrapped up soon with the need to do wetlands delineation identified.

On 8-16-2023, Carl Stevenson emailed aeration basin white sudsy foam videos. On 8-18-2023, Gary Cohen emailed that EPA will be making a judicial referral to the US DOJ for the Maxson Facility for on-going SNC violations over the last 1 ½ years. They do not believe an Administrative Order is sufficient due to the number of violations. They were not aware that most of the violations were caused by construction and would continue as construction continued.

The Maxson Facility got a 3rd blower to start on 8-21-2023 after power blip after a week of low D.O.s. The Maxson Facility received NOV QNCR for Maxson from TDEC for May and July 2023 on 8-25-2023. Meanwhile, Chris Luss began coordinating with Biorem to get Bioreactors 3 & 4 ready when Lagoon 2A project is completed (new lagoon cover and pumps).

In September, the need to plan for adding gates for isolating the Maxson Facility CBS in the future was identified. This is due 1) reduced number of screens from 3 to 2, 2) deterioration of metal parts due to hydrogen sulfide, and 3) fragile nature of the rake mechanism. This work could be performed by use of the 60-inch bypass or bypass pumping and stop logs.

Maxson facility influent sampling appears to be more representative this summer after the collapse of the sewer. The collapse appears to be holding back a lot of extraneous solids that tending to get into the sample and bias it high. When this factor is accounted for, the normal sampling procedure is more reflective of industrial user SMR and expected domestic loading.

The new Data Center should be ready next month. Change over to new headworks odor control was planned involve disconnecting original odor control likely over 2-3 week period. There was delay on Solar Farm due to MLGW wanting to expand for the grant.

On 9-20-2023, Material Matters stated 10-15 mg/kg was guideline for molybdenum for industrial impact. Molybate was identified as common in closed loop system according to a WEF Cooling Tower webinar on 9-21-2023.

On 9-22-2023, the Judge issued the final ruling regarding HLSD lawsuit. The jurisdiction has until the end of the year to submit a schedule for the construction of both peak flow reduction facility and wastewater treatment facilities and progress reports submitted every 6-months thereafter for the docket. Rates will escalate every year until 7-1-2031 when they will equal the Volumetric Rate. Beginning 10-1-2033, HLSD would pay the greater of the volumetric rate or applicable sewer rate in effect. Primary Clarifier #2 was taken down for repair as the concrete floor had been busting up and getting into bio-pumps.

In October, Xylem continued fusing HPDE pipe for the bypass and constructed activities generally hindered by high hydrogen sulfide levels (H2S > 200 ppm). All 4 secondary clarifiers on north side were taken out of service as well as north aeration basins to work on return sludge pumps. Maxson Facility ran on ½ the plant for to repair the north side return sludge pumps for a good bit of the month with foaming due likely to either young sludge or overloading even though process liked to blame industries.

Additions of clarifiers, scum pumps, and RAS pump status to Ovation screens were delayed until the new DCS was installed in new Admin Bldg. However, as previously mentioned, the integration of the old and new DCS as well as the connection of new unit processes proved to be problematic due to lack of project direction.

Upcoming discussion with TVA were planned to use treated plant effluent and get off potable water. The question is who would own and operate the effluent re-use system. Possibly the Maxson Plant could use some of the tertiary treated water. TVA could also send treated water back to the Maxson Plant which would eliminate the need for the effluent strainer.

CBS roller bearings were observed corroded and would need to be upgraded with stainless steel bearings sold by Vulcan. Coated bushings are also available. The Stiles Facility has the same problem.

Mississippi River stage was at -12.04 feet on 10-17-2023, lowest on record. In 2022, it was -10.81 feet on 10-21-2022, the second lowest on record. On 10-22-2023, 2 RAS pumps on North Side of the Maxson Facility became functional and the treatment returned to two-sided operation. Xylem continued cleaning 13/14 North Aeration Basin FBD using staff from the pump bypass. A vendor began putting a CCTV camera in the 60-inch bypass after pumping out the levee structure to evaluate if the 60-inch bypass could be used as such.

Meanwhile, on 10-30-2024, Chris Hill Construction crews continued to make efforts to cut into the top of the pipe to install the withdrawal tubes for the Nonconnah Interceptor Bypass as the Maxson Facility increased pumping to maintain lower sewer levels and as well as installing new CBS rakes where the exact opposite, backing up the sewer, would have been ideal.

Emerson installed new DCS at the Maxson Facility the week of 11-6-2023 now that the Purafill system was installed. However, much of the original data would be erased. Xylem still has not

started the bypass due to problem cutting into T-lock liner. In the meanwhile, at the construction site, H2S levels in the 200+ range were occurring in the afternoons. Solae began sending out dryer notices again.

In November, the City met with TVA about tertiary treatment of Maxson Facility effluent for cooling water. The system would be operated by either Stantec or Ovelia. The next meeting with TVA was scheduled in February 2024 to review TVA taking effluent. Xylem cleaned FBD in 1 basin (north) and DWP reduced to the low 20s. CDM continued waiting on proposal from subcontractor to repair aeration basin header leaks. Chris Hill installed CBS parts on 11-17-2023. The Maxson Facility received QNCR from TDEC on 11-21-2023 for Aug-Sep-Oct 2023 violations. Chris Hill finally was able to install withdrawal tubes of Nonconnah 96-inch interceptor on 11-28-2023.

Maxson Facility unmonitored BOD, COD, and TSS as presented in the monthly mass balance report began tightening up in May 2023 ending the year at close to expected values. Riviana started a new rice cooking line. The 22KV and Disinfection IPA HVAC units were replaced 12-5 to 12-6-2023. The new RAS pumps tested ok after vibration test.

The Maintenance Department completed numerous noteworthy tasks during the year.

In January, crews installed and put into service temporary ABF pump #3. In February, the defective 600-amp breaker located at the Pump Building that feeds the maintenance shop building power was replaced. Crews installed and put into service temporary ABF pump #5.

In March, the fine bar screen rake assembly was rebuilt, installed, and put back into service at #2 Fine Bar Screen. At #2 South Return Sludge Pump, a return sludge pump was rebuilt, installed, and put back into service. All 5 corroded galvanized louver panels were replaced with aluminum panels on the south side of the Secondary Dewatering Building.

In April, the Storeroom was re-roofed to leaks. Facia boards were replaced and gutters provided. A new 100 Mill HDPE cover on a carbon tubular frame and aluminum alloy steel was re-engineered to replace deteriorating Grit Tank # 2 and 3 odor control cover. The Blower Building Filter deteriorated blower filter room doors were replaced and conventional locks upgraded to barrel type locks. A new 100 Mill HDPE cover on a carbon tubular frame and all aluminum alloy steel was re-engineered to replace deteriorating Grit Tank # 1 and 4 odor control cover.

In May, fabrication and installation of all four aluminum grit tank covers and cover mover was completed at the Grit Tanks. The #1 South Coarse Bar Screen rake assembly was rebuilt, installed, and put back into service. The East lift Station Pumping was re-engineered from a stationary centrifugal pump set up to submersible pumps on a rail and one well set up. The #2 North Return Sludge Pump was rebuilt, installed, and put back into service. The Biorem Heater # 3 was replaced with a Tankless Heater and busted copper tubing due to freezing temperature was replaced and pipping insulated.

In June, at 2 North Blower, all blower expansion joint bolts were replaced and/or tightened. The old vacuum pod on header was removed and new unit installed at #10 ABF Pump. The return sludge pump was rebuilt and installed at #3 South Return Sludge Pump and put back into service. A new HDPE water supply line was run to 2 North Blower. The temporary line was

disconnected, and a new HDPE supply line connected. The old vacuum pod on header was removed and new unit installed #2 ABF Pump. The #1 ABF tower arms were balanced and tightened brackets to make tower turn. The #2 Effluent Water Pump was rebuilt and installed. The burst Biorem Recirculation Pump caused by minus freezing temperatures was replaced and pipes insulated. The #7 Screw Pump broken bolts were extracted, the rotating unit rebuilt, reinstalled, and put back into service.

In July, all of #1 North Blower bag filters were changed out and the filter room was cleaned up and washed down. All of #1 South Blower bag filters were changed out and the filter room was cleaned up and washed down. All of #3 North Blower bag filters were changed out and the filter room was cleaned up and washed down. The Lagoon Used Oil Storage Tanks were primed and painted. The #3 Fine Bar Screen worn screen rake was removed and reburnished assembly installed. The failed upper rotating unit of #2 Screw Pump was removed and refurbished unit installed. The #2 North Blower was repaired, the oil line rebuilt, the auxiliary oil pump was rebuilt, and both pump and motor couplings replaced. A new HDPE water supply line was run to #1 South Blower and the temporary line disconnected. The #2 Fine Bar Screen drive bearings were changed out, new gasket seal material installed on all 3 press drain pans, rack rollers were replaced, rotary and union for press water supply line replaced. A Screw Pump Flight was sandblasted, primed and entire screw assembly painted. The #4 Effluent Water Pump rebuild was completed and installed.

In August, the #4 Raw Sewage Motor was removed, refurbished, and reinstalled, including all connections and alignments. A new overhead door and operator were installed at the Plant Maintenance Shop. Two 2 new compressors and filter dryers on HVAC unit for the Plant Maintenance Shop were installed. All #2 North Blower bag filters were changed out and filter room cleaned up and washed down. All #2 South Blower bag filters were changed out and filter room cleaned up and washed down. All #3 South Blower bag filters were changed out and filter room cleaned up and washed down. A spare Screw Pump upper rotating was rebuilt and installed on #7 Screw Pump. The old inoperable actuator was removed and new actuator installed on #6 ABF Pump discharge line. The old inoperable actuator was removed and new actuator installed on #9 ABF Pump discharge line. The sweep arm brackets #2 North & 3 South Clarifiers were heated, straightened, and sweeps installed on each. The old rusted diesel tank at the Lagoon was demolished, a concrete foundation poured, and new double walled 4000 gallon tank and spill tub were installed. A spare Raw Sewage spare raw sewage shaft was fabricated, including keys and coupling locking nut.

In September, the old external oil pump and motor of #1 South Blower was removed and new units installed. The South Coarse Bar Screen drive shaft was cut to remove locked rake assembly, refurbished and rack assembly reinstalled rake and monitoring of it performance begun. The #7 ABF Pump old discharge actuator and spline gear were removed and new spline gear and actuator installed. The #8 ABF Pump old discharge actuator and spline gear were removed and new spline gear and actuator installed. The North Coarse Bar Screen pin rack rollers and bushings were changed out, adjustments made, and put back into service. The #7 Screw Pump failed gear box was removed, a complete rebuild performed, reinstalled, aligned, all connections made, and put back into service. The South Coarse Bar Screen Rake Assembly was refurbished and whole bar screen Rake Assembly replaced, having been stuck 10 feet below the top floor grade level. A 20-feet drain box was set between #110 North and #110 South Lagoons. A dewatering box west of Scum Pit was set up.

In October, all #2 North Blower old oil-soaked bag filters were removed, new bag filters installed, and oil pump motor replaced. A new 6-inch plug valve in the water line on north side of Return Sludge was removed and installed. A contractor punctured the new 16-inch HDPE primary that caused spillage. Primary Sludge system was temporarily shut down and pipe repaired. The North Coarse Bar screen pin rack roller and bushing were replaced. The #1 North Return Sludge Pump was removed and refurbished pump installed, valve repaired, pump and motor couplings fabricated, and aligned. The South Coarse Bar Screen pin rack rollers and bushings were changed out, adjustments made, and put back into service. The #3 Fine Bar Screen pin rack rollers and bushings were changed out, adjustments made, and put back into service. The 24-inch swing check valve of #3 North Return Sludge Pump was removed and new valve installed. The North Side Return Sludge seal water supply line header to each pump on the north side completely re-piped. The broken 12-inch drainpipe in #3 Field was located and replaced west of #3 Field to use as surface disposal site for biosolids from the Stiles Facility. The filled in #2 manhole was built up for levee gate for 60-inch plant bypass line.

In November, the #3 Fine Bar Screen Rake assembly was refurbished, pin rack rollers and bushings replaced, and rake assembly reinstalled, and put back into service. The South Side Return Sludge seal water supply line header to each pump on the north side was completely re-piped. The problem #4 Raw Sewage Motor was removed, refurbished, reinstalled, connected, aligned, and put into service. The snapped South Coarse Bar Screen power cable was replaced.

A new 12-inch hydraulic power unit was set up from #5 Lagoon to the east side of the #110 South and from #5 Lagoon to #110 South.

In December, #1 Raw Sewage Motor and Mag were removed and completely refurbished. A fine bar screen rake was refurbished with new gearbox, new drive shaft, new guide arms, new bearings, new rollers, and new cogwheels as spare unit.

In addition to the major tasks listed above, the Maintenance Department completed 5733 Preventative Maintenance Work Orders, 242 Scheduled Maintenance Work Orders, and 298 Unscheduled Maintenance Work Orders for a total of 6273 work orders. The annual number of work orders completed over the last 10 years is summarized as follows:

Year	2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
Preventative	5733	4967	5067	7435	3514	3417	1896	178	567	1000
Scheduled	242	175	558	1153	871	799	321	20	72	96
Unscheduled	298	304	622	1083	1756	1190	252	18	118	125
Total	6273	5446	6247	9671	6141	5416	2469	265	772	1234

Ferrous Chloride Use –No ferrous chloride has been used at the Maxson Facility since January 2012. Among its varied uses, ferrous chloride reduces the likelihood of struvite formation by producing acidic conditions and binding up magnesium and phosphorus ions. Struvite formation began to be observed in Lagoon Nos. 3 and 4 in 2015, favoring areas where there is turbulence. A 5000-gallon tank has been placed into position to add ferrous chloride to feed sludge prior to dewatering when this practice is resumed. No ferrous chloride was purchased or used in 2022. In

the future, purchase options would include commercially made ferrous chloride containing about 34% or spent pickle liquor containing 10% iron.

Flow – The average daily flow rate in 2023 was 66 mgd which was 2 mgd lower than the previous year. The 0.5th percentile flow in 2023 was 48 mgd which is considered base flow for the year. The average daily flow received from the major significant industrial users was 9.2 mgd. This leaves 8.8 mgd for 87% of average daily flow accounted for. The total rainfall for the year was 55 inches (NWS). Memphis annual rainfall has averaged 54 inches.

BOD_5 – The average influent BOD_5 loading for 2023 was 388,482 ppd day, which is 74,494 pounds or 16% decrease from the previous year. The average influent BOD_5 concentration was 702 mg/l and the average effluent concentration was 116 mg/l, resulting in a removal efficiency of 83%. The total average BOD_5 reported being discharged by the major significant industrial users was 219,125 ppd in 2023, equivalent to 56% of the total loading. This is 71,578 pounds or 25% less in monitored industrial loading from the previous year. The 0.5 percentile loading received in 2023 was 123,901 ppd which is the assumed domestic loading for this year. This leaves 45,456 ppd of influent BOD_5 for 88% of average influent BOD loading accounted for.

Total Suspended Solids (TSS) – The influent TSS averaged 331,840 ppd, which is 85,068 pounds or a 20% decrease from the previous year. The average influent TSS concentration was 587 mg/l and the average effluent concentration was 145 mg/l resulting in a removal efficiency of 75%. The total average TSS reported discharged by the major significant industrial users was 124,480 ppd in 2023, equivalent to 38% of the total loading. This is 23,480 pounds or 16% decrease in monitored industrial loading from the previous year. The 1 percentile loading for 2023 was 105,390 ppd which is expected domestic loading for this year. Lagoon supernatant and dewatering filtrate return averaged 3,783 ppd. This leaves 98,187 ppd of influent TSS for 70.4 % of average influent TSS loading accounted for.

Chemical Oxygen Demand (COD) - The influent averaged 702 mg/l and 827,361 ppd, which is 218,827 pounds or 21% decrease from the previous year. The effluent averaged 366 mg/l and 201,461 ppd. The average removal was 48%. The 1 percentile loading for 2023 was 214,327 ppd which is expected domestic loading for this year. The total average COD discharged by the major significant industrial users was 434,996 ppd which equals 53% of the total loading. This leaves 178,038 ppd for 78% of influent COD accounted for. The COD surcharge is based on covering 50% of disinfection costs.

Effluent E. coli - E. coli analysis was performed on daily effluent grab samples throughout the year as required by the NPDES permit as report only. Effluent E. coli results for this past year are presented in Figure 1 showing the periods of initial and final startup of disinfection.

Water – In 2023 metered water usage was 56,023 CCF per month resulting in total usage of 672,273 CCF or 1.4 MGD. The total annual cost was $881,454. This is a decrease of 21,776 CCF per month from the previous year.

Energy – The main power consumption of the plant was an average of 3,965,107 kwh per month in 2023. This is 639,141 less per month than the previous year. The monthly peak demand averaged 6406 kW which is 746 kW less than the previous year. The average monthly main power bill was $$335,911, which is $57,433 less than the previous year. Payments for

participating in the ENERNOC curtailment program for main power are unavailable but are likely minimal due to construction activity and equipment problems.

January was the high month for demand at 7168 kw and the high month for consumption at 4,678,607 kwh. September was the low month for demand at 5479 kw and December was the low month for consumption at 3,462,602 kwh. Daily electrical demand throughout 2023 is shown in Figure 2.

Effective 10-1-2015, the City entered into an agreement with MLGW to switch from Seasonal Demand to Time of Use billing. TVA would charge Fuel Cost Adjustment based on user class. TOU rate structure benefits high load factor customers through incentivizing off-peak energy usage. The contract term is 5 years and will be automatically extended unless otherwise terminated.

Sludge and Biosolids – The quantity of primary sludge wasted to the lagoons in 2023 was 49.8 million pounds, a decrease of 54.2 million pounds from the previous year. The quantity of secondary sludge wasted to the lagoons in 2023 was 58.7 million pounds, a decrease of 10.3 million pounds from the previous year. The total quantity of both primary and secondary sludge wasted from the plant was 107.5 million pounds, a decrease of 44.9 million pounds from the previous year.

Approximately 2.8 million pounds of digested sludge was dewatered at the secondary dewatering facility and disposed in the 800 acres available for Surface Disposal, of which 309 acres is considered permanently disposed. This includes approximately 40.6 million pounds of iron from previous years that was added to Lagoon Nos. 3, 4, and 5 (the uncovered lagoons). Approximately 24% of the combined primary and secondary sludge wasted to the lagoons in 2023 was destroyed equivalent to 25.5 million pounds. The formula used to calculate percent destroyed is %VS of wasted primary and activated sludge multiplied by %VAR. The %VS for primary and waste activated sludge is based on two separate sampling points and then combined mathematically based on proportion. The %VS for feed sludge is based on a single sampling point.

Considering removal and volatilization the new change in the lagoon solids inventory was an increase of 78.6 million dry pounds within the approximately 257 acres of covered and open lagoons. The calculated cumulative dry solids within the covered and uncovered lagoons is 625.8 million dry pounds. The calculated dry bio-solids that has been surface disposed since 1979 is 1508 million dry pounds.

A total of 2.8 million pounds of digested sludge was disposed in 2023 as surface disposed biosolids (1400 tons) and volatilized lagoon sludge (12,750 tons) at an estimated total cost of $4,873,291.

The annual cost per dry ton for digestion, dewatering, surface disposal and volatilization for the past 10 years is as follows:

2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
$344	$98	$108	$74	$112	$67	$66	$76	$68	$68

The annual cost per dry ton for digestion, dewatering and surface disposal for the past 10 years is as follows:

2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
$3,481	$406	$468	$291	$549	$265	$222	$201	$214	$262

The annual amount of digested sludge (in million dry pounds) removed from the lagoons over the last 10 years is provided below:

2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
2.8	21.7	20.4	30.7	15.8	23	24.7	29.95	27.2	23.4

For the last 10 years, net accumulation has been occurring at an average rate of 39 million dry pounds per year. Surface disposal has averaged 22 million dry pounds per year. To stop net accumulation, the Maxson Facility will need to de-water and surface dispose an average of 61 million dry pounds per year which is 22 times the existing rate and would be 96 percentile level of performance. Hopefully, the upcoming project to clean out, subdivide, and cover Lagoon 5 will correct this situation.

Biogas – For the year 2023, the Maxson Facility produced a total of 1 million cubic feet of biogas. TVA used zero million cubic feet of the annual production.

The annual biogas produced by the Maxson Facility for the past 10 years is as follows (in millions of cubic feet):

2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
1	191	430	438	552	565	744	1167	1254	1155

The annual daily average biogas produced for the past 10 years is as follows (in millions of cubic feet):

2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
0.0	0.52	1.2	1.2	1.51	1.55	2.04	3.20	3.44	3.16

Annual biogas yield for the last 10 years (ratio of biogas to volatiles destroyed) in cubic feet per pound is as follows:

2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
0.04	2.8	6.3	4.9	8.9	8.3	12.4	23.8	21.4	14.1

The amount of sludge volatilized each year is relatively constant with no discernable long-term trends or impact on the volume of biogas produced. What is variable is the amount of biogas produced per pound of sludge destroyed as shown above. Since 2017, biogas volume has dropped to levels not seen since 2003 before Lagoons 2A, 2B, and 2C were first covered. Trends in biogas production over the years are shown in Figure 3.

SCHD issued the final Title V permit on 10-4-2019, Permit # 00701-01TV, expiring 10-4-2024. A new permit application would be required by 4-27-2024.

The August-December 2022 Semi-Annual Report was transmitted to the SCHD in February. SCHD performed annual inspection on 3-23-2023. The January-July 2023 Semi-Annual Report was transmitted to the SCHD in August.

The annual biogas usage by TVA along with the amount of biogas flared as waste gas by the Maxson Facility over the last 10 years is as follows (in millions of cubic feet):

Year	2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
TVA	0	0	0	0	0	0	0	0	0	60
Flared	1	191	430	438	552	565	744	1167	1254	1094

The City received zero in revenue from TVA for the Maxson biogas for 2023.

Permit Violations – This year the Maxson Facility had 803 excursions of its NPDES permit limits related to treatment efficiency and performance. The number of excursions per month was steady throughout the year and a function of the number of trickling filters in service, ability of the blowers to meet air demand, and changes in influent loading due to industrial activity. There were typically 2-3 trickling filters and 2-3 blowers in service each month and effluent quality reflected this. All fine bubble diffusers were in service by the end of March. The excursions are described as follows:

In January, there were 77 permit excursions which was an increase from 74 the previous month. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 4 weekly pounds, and 28 daily concentration excursions. There were TSS monthly average concentration, monthly average pounds, 3 weekly average concentration, 3 weekly average pounds, and 15 daily concentration excursions. There were 15 effluent dissolved oxygen excursions.

In February, there were 76 permit excursions which was a decrease from 77 the previous month. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 4 weekly average pounds, and 20 daily concentration excursions. There were TSS monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 4 weekly average pounds, and 18 daily concentration excursions. There were 16 effluent dissolved oxygen excursions.

In March, there were 83 permit excursions which is an increase from 76 the previous month. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 3 weekly average concentration, 3 weekly average pounds, 20 daily concentration, and 1 daily percent removal excursions. There were TSS monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 4 weekly average pounds, 20 daily concentration, and 5 daily percent removal excursions. There were 1 effluent settleable solids and 16 effluent dissolved oxygen excursions.

In April, there were 85 permit excursions which is an increase from 83 the previous month. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 5 weekly average concentration, 4 weekly average pounds, and 19 daily concentration excursions. There were TSS monthly average concentration, monthly average pounds, monthly average percent removal, 5 weekly average concentration, 4 weekly average pounds, 19 daily concentration, and 4 daily percent removal excursions. There were 6 settleable solids and 13 dissolved oxygen daily effluent excursions.

In May, there were 74 permit excursions which is a decrease from 85 the previous month. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 3 weekly average concentration, 2 weekly average pounds, and 13 daily concentration excursions. There were TSS monthly average concentration, monthly average pounds, monthly average percent removal, 3 weekly average concentration, 2 weekly average pounds, 15 daily concentration, and 8 daily percent removal excursions. There were 7 settleable solids, and 15 dissolved oxygen excursions.

In June, there were no permit excursions which is a decrease from 74 the previous month. This was due to Solae being down for 19 days and the remaining days at either ¼ or ½ production. This resulted in a sampled average influent BOD loading of 264,738 ppd versus an expected loading of 260,327 ppd and a sampled influent TSS loading of 198,871 ppd versus an expected loading of 168,593 ppd. This is a good base line of influent loading conditions during an extended period of curtailed production from Solae.

In July, there were 88 permit excursions which is an increase from zero the previous month. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 3 weekly average concentration,1 weekly average pounds, and 11 daily concentration excursions. There were TSS monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 2 weekly average pounds, 17 daily concentration, and 5 daily percent removal excursions. There were 19 settleable solids, 19 dissolved oxygen, and 1 pH daily effluent excursions.

In August, there were 79 permit excursions which is a decrease from 8 the previous month. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 3 weekly average pounds, and 13 daily concentration excursions. There were TSS monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 2 weekly average pounds, 13 daily concentration, and 5 daily percent removal excursions. There were 6 effluent settleable solids, and 23 effluent dissolved oxygen excursions.

In September, there were 67 permit excursions which is a decrease from 79 the previous month. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 5 weekly average concentration, 2 weekly average pounds, and 14 daily concentration excursions. There were TSS monthly average concentration, monthly average pounds, monthly average percent removal, 5 weekly average concentration, 3 weekly average pounds, 14 daily concentration excursions and 3 daily percent removal excursions. There were 4 effluent settleable solids and 11 effluent dissolved oxygen excursions.

In October, there were 83 permit excursions which is an increase from 67 the previous month before. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 3 weekly average pounds, and 19 daily concentration excursions. There were TSS monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 1 weekly average pounds, 17 daily concentration, and 3 daily percent removal excursions. There were 6 settleable solids, 19 dissolved oxygen, and 1 pH daily effluent excursions.

In November, there were 76 permit excursions which is a decrease from 86 the previous month. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 3 weekly average pounds, and 16 daily concentration excursions. There were TSS monthly average concentration, monthly average pounds, monthly average percent removal, 3 weekly average concentration, 3 weekly average pounds, 17 daily concentration, and 2 daily percent removal excursions. There were 14 effluent settleable solids and 8 effluent dissolved oxygen excursions.

In December, there were 95 permit excursions which is an increase from 76 the previous month before. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 5 weekly average concentration, 4 weekly average pounds, and 21 daily concentration excursions. There were TSS monthly average concentration, monthly average pounds, 5 weekly average concentration, 5 weekly average pounds, 22 daily concentration, and 3 daily percent removal excursions. There were 14 dissolved oxygen and 10 daily settleable solids effluent excursions. The excursions were concentrated through 12-25-2023 after which compliance was achieved. The increase in excursions compared to other months during similar conditions could not be readily determined.

The cumulative violations are shown in Figure 4 with Stiles Facility cumulative violations for comparison. The excursions associated with different major periods of plant history are summarized below:

Period	Average Number of Excursions Per Year	Description
1982-1985	30.5	First plant capacity expansion (Consent Decree)
1986-1994	4.0	Industrial loading increasing toward full capacity
1995-1997	24	2nd plant capacity expansion (Voluntary)
1998-2010	4.4	Steady loading conditions
2011-2020	51.5	Collection system and plant equipment failures; subsequent plant equipment replacement projects
2021-2023	939.7	3rd plant capacity expansion (Voluntary)

The Maxson Facility has had two years (1987 and 2010) with zero violations. The highest number was 1524 violations in the year 2022.

During 2011-2023, 353 of 3334 violations at the Maxson Facility, or 10.6%, were for "pounds" permit limit excursions which the Stiles Plant does not have to report, creating regulatory imbalance between the treatment plants. The violations per year on a percentile basis since 1982 are summarized below:

Percentile	1982-2020	1982-2023
0.25	3	3
0.5	8	10
0.6	15	21
0.625	15	23
0.65	23	24
0.7	24	27
0.75	27	31
0.8	30	48
0.85	44	65
0.9	56	125
0.94	96	610
0.95	112	812
0.96	119	1062
0.97	126	1338

The above table shows the impact the of the 2021-2023 violations on the 40-year dataset starting at the 65th percentile level.

Disinfection, Process Improvements, and Odor Control

In 2014, the disinfection, process capacity and treatment improvements, and odor control projects were combined into a single initiative called Construction Manager at Risk (CMAR). Treatment capacity increase was based on an average daily flow of 90 mgd at influent BOD concentration of 800 mg/l equivalent to 600,000 ppd as well as compliance with 30-30 BOD and TSS performance standards.

Package 2A / GMP 3 covering construction of two new secondary clarifiers, intermediate pumping improvements, odor control, and fine bubble diffusers continued from the previous year. In March, all aeration basin FBD were finally installed but Aeration Basin 14/15 North could not be utilized until a massive air header leak repaired. Square D replaced 3 HVAC compressors for the new 23KV IPA. City had been previously using rental units. In June, the new 5N/5S clarifier floor was observed to be cracking up due to faulty levelling grout. In July, it was determined that since the flow control structure work was tied to construction on TF6 that the contract may have to extend at least another year. As of October, the odor control project was delayed due to the need for new pad-mounted switch gear to power the head works odor control with a lead time of 1-2 years. Meanwhile, CDM continued waiting on the new

transformer to complete Odor control which should complete GMP 3 in April 2024. The two new secondary clarifiers were put into service on 12-1-2023 after repairs had been made.

Package 2B / GMP 4 for constructing re-aeration tanks and a new blower building continued from the previous year. In April, CDM completed installation of FBD in RAS Basins 1,2,3, 4 and began waiting on delivery of the 3 RAS Blowers.

Package 3 / GMP 5 for constructing and installing new bio-tower media, odor dispersion fans, and tower improvements continued from the previous year. TF5 started on 30-day test period on 1-18-2023 including a 10-day startup for distribution arms. Initial rotation was 1 reviolation per 9 minutes. Plans are to have three new working TFs within next 7-8 months. The three existing TF (1, 2, 3), however, would remain in service for the rest of the year as described later. The startup of TF5 continued throughout February. Pan tests were performed to confirm TF5 distribution arms were properly working and had to be repeated due to debris clogging the distribution arms. Work continued in March with TF5 start-up with expectations to be in full service by 5-1-2023. The wait continued for parts needed for TF6 startup which was already stacked with media. In March, TF5 continued to fail distribution arm testing with scattered spray pattern resulting in a need to install different orifice sizes. The proper distribution of flow is very critical because towers have a lot more media and need to have equally distributed coverage. TF4 media began arriving toward the end of the month as Gulf Coast Underground continued coating TF4 in preparation for stacking. In April, TF5 finally passed the pan test. However, temporary HVAC for compressor building for turning TF5 distribution arm was required since Brentwood was not aware of requirement to coat HVAC gear. TF4 may have more issues with installation of coating, possibly related to humidity. During May, concern was expressed about compressor used to drive distribution arm and coating coming off effluent trough for TF5. In June, the commissioning of TF5 and the coating of TF4 continued to be held up due to need to recoat TF5 and TF6 troughs. A coating consultant was contacted to take samples of both walls and troughs.

In July, the need to remove coating on TF5 due to on-going failures of coating was evaluated. In August, plant still operated on 2-3 TF due to continued dithering over what to do about the bad coating on TF5 & 6. The timing of TF construction was re-sequenced to allow 3 TF to be on line at all times. The final coating report on 8-15-2023. Alumina Liner installed at TF4, 5, and 6. Surface preparation was deficient, too rough and the coating allowed to cure too quickly. Curing process not done per manufacturer's specifications. Gulf Coast was subcontractor. TF4 & TF5 coating needed to be removed and redone. TF6 expected to be start after 8-27-2023 with commissioning of TF6 to wait until summer of 2024. In the meanwhile, TF1 media continued to be received and installation of the leveling grout on 5N and 5S clarifiers was completed. As of October, no removal of the recently installed new media was performed on TF5 and TF6 pending 3rd party testing of coating. However, removing coating on TF4 and TF6 troughs continued. Magnolia continuing to install concrete supports for FRP odor control duct. The plugs in TF4 and TF5 discharge pipes were re-installed to prevent backup into effluent troughs during removal of calcium aluminate. In November, Gulf Coast Underground completed removal of calcium aluminate from troughs, outside walls, and window openings of TF 4,5,6. In December, coating issue continued to delay startup of TF5 and TF6 as a decision to remove the recently installed media from TF5 and TF6 had not been made. The current assumption is that it will have to be removed and the coating removed and re-applied. The vendor was on-site to perform surface prep of trough area on TF4 and moved on to work on TF6.

WIFIA Projects

The EPA announced the closing on the WIFIA loan on 10-8-2020 for $156 million which kicked off the start up of the new Admin Building, Lagoon 2A remediation, Primary Pump Station Upgrade, Secondary Dewatering Upgrade and Lagoon 5 expansion projects.

Power was run to the new Admin Building and work on the back parking lot began in March. Power was brought in beginning April. The new DCS was installed once HVAC and Purafil systems were in place. Emerson completed terminating wires for the new Data Center in August. The new Admin Building was completed 12-8-2023 with issues with the Fire Alarm and the DCS remaining.

In January, the NTP for the Lagoon 2A Phase 2 project was given to order major equipment and materials. The project was held up regarding the existing ring wall structure. Evaluation of the structure was initiated and completed by about 10-5-2023. Demolition of pumps and motors started in July. In November, final changes made in ring wall design. Construction of the pump building began in December.

A pre-bid meeting for the Primary Pump Station Upgrade was held on 3-2-2023 with a bid date on 7-12-2023. Bids received 8-9-2023 and Chris Hill Construction was later awarded the contract. As of December, the project had not started.

A fully executed contract with Chris Hill Construction for the Secondary Dewatering Upgrade was received 5-24-2023 with construction tentatively to start 6-19-2023 with completion by 12-10-2024. NTP issued in November.

Lagoon No. 5 Expansion is at 100% design and is scheduled to be initiated once the other projects are complete. Kick-ff meeting with B&V was on 8-8-2023 to review design and prepare for bidding.

Biogas Quality Investigation

In 2023, Hydrogen Sulfide concentrations averaged 105 ppmv based on one sample due to limited biogas production. Energy values averaged 388 BTU/cubic feet (wet) based on one sample due to limited biogas production. Average annual values of analytical results from previous years are summarized as follows:

Year	Soloxanes (ppbv)	Hydrogen Sulfide (ppmv)	BTUs/cubic foot (wet)
2013	124	19,820	468
2014	156	4591	441
2015	152	2836	483
2016	NA	935	320
2017	NA	1981	420
2018	NA	4540	350
2019	NA	1075	410
2020	NA	2012	362
2021	NA	2364	436

2022	NA	2433	253
2023	NA	105	388

TVA Conversion to Natural Gas Related Projects

In August 2017, installation of the 5 new flares was completed by Kiewit Construction under TVA's contract and flaring of the waste gas using the new flares initiated. In September 2017, installation of the 4 bioreactors was completed and the reactors charged with media. Kiewit's contract to complete the job ended 11-5-2017 and operation turned over to the City. The City continued to work with CDM Smith and Biorem to work out warranty issues related to construction as well as Biorem's obligation to achieve steady-state operation so that maximum hydrogen sulfide levels do not exceed 480 ppm per the TVA contract with the City. On 12-12-2021, Biorem completed their performance trial. The system was able to meet contract specifications with untreated biogas H2S concentrations up to 10,000 ppm.

The revenue contract was signed with TVA on 1-3-2018 with a 20-year term effective once biogas treatment is adequate. The estimated capital contribution is $20 million based on a 6,500,000 MMBtu minimum agreement volume with TVA concurrently agreeing to pay the City 33 cents per MMBtu spread out over a 13-year payback period. This is based on a minimum annual average 500 BTU wet (higher heating value) and maximum monthly average Hydrogen Sulfide concentration of 30 grains per 100 CF or 480 ppm.

The City issued a Notice of Transfer of Biogas to TVA to begin 2-1-2022 and began sending production data to TVA semi-monthly. Biogas flow meters and totalizers were installed at the beginning of 2023. Bar Environmental was contacted to install new heat exchangers that were damaged during the January freeze. The need was identified to purchase spare heat exchangers. A limited amount of biogas was treated and flared in May. In June, only 2 of the 4 reactors were functional. It was observed that one set of spray nozzles had been broken off, possibly due to media addition. The biogas analyzer was down due filters saturated with water since the insulation on the sampling tubes had degenerated. In July, the Biorem was in operational status. Late in the year the Biorem was system drained to prevent freezing. Recirculation pumps were used to keep the reactors in service when no biogas was available.

TVA continued to dewater pore water and stormwater from the east ash pond and discharging it to outfall 002 which goes to the cooling channel. Construction continued throughout the year for the groundwater remediation system. Equipment and piping tests and extraction well testing was performed. Startup of discharging treated ground water to the POTW began 3-10-2023. Ash continued to be removed from the west ash pond and backfilled. Total load out through August 2023 was 470,600 CY to the landfill with the project completed in September 2023. Ash was removed from the east ash pond and backfilled. Total load out through August 2023 was 256,800 CY to the landfill with completion expected in November 2029. The 36-inch bypass to MH23 was put in service on 9-14-2022. The permanent sewer line is expected to be completed in October 2024. Deconstruction/demolition of the Allen Fossil Facility began in about August 2022 with expected completion in 2025.

Biosolids Long-Term Planning

The Maxson Facility began operations in March 1975. Lagoon #1 was constructed along with the main plant. Lagoon #2 was then constructed as a supernatant clarifier in 1977-78 with

supernatant returning to the head of the plant along with Lagoons #3, #4, and #5. TDHE approved the 160 acres south of Lagoon #1 for lime stabilization. The City sent a draft interim sludge handling plan to TDHE on 5-11-1978. The Consent Decree entered on 7-28-1978 had a section requiring the City to initiate an interim solution for "sludge handling" facilities. Pursuant to the Decree, the City built an 8-acre pond (Lagoon #3) and proposed to spray-irrigate the sludge on an adjoining field in the dry weather months. On 12-1-1978, the City began discharging sludge to Lagoon #3 which was anticipated to provide storage capacity for one year. As of 4-10-1979, the lagoon was at capacity and sludge began to flow back through the plant. The Second Amendment to the Consent Decree entered on 5-7-1979 allowed the City to intermittently discharge sludge through the plant's bypass until there was sufficient capacity to store sludge until startup of the land application program later that year. A revised sludge handling plan was submitted to TDHE on 7-2-1979. This plan stated that lime stabilized sludge would be applied to the 110 acres just south of the anaerobic lagoon during summer of 1979 with approximately 350 acres of additional land available adjacent this property for future use. On 1-2-1980, the City submitted a final revision of the plan that focused on sludge land application. By 5-6-1980, the City received approval for a sludge spray irrigation system on the 110 acres (west of the lagoons) with an additional 310 acres available for future use. The City acquired two sludge trucks with 4000-gallon tanks in May of 1980 and received a permit to apply sludge to a 310-acre dedicated site adjacent to the lagoons in June of 1980. By that point, a total of 280 million gallons of sludge storage capacity was available for stabilization and further digestion, followed by application on approximately 420 acres of dedicated land. On 4-21-1981, the Director of Public Works informed the City Council that the City had 608 days of sludge storage capacity left at the Maxson Facility and 375 days left at the Stiles Facility.

Sludge reports submitted to TDHE from 1980 to 1992 documented quantity of sludge applied and analytical results and as time went on, surface water run-off analytical results. Four sludge application sites were approved for the Maxson Facility beginning with Site 1 (110 acres) in 1979, followed by Site 2 (310 acres), in 1980, Site 3 (180 acres) in 1983, and Site 4 (3500+ acres) in 1982. Most of the sludge was applied to Sites 1, 2 and 3 at a rate of 15 to 20 dry tons per acre per year in an area that became known as the Earth Complex. Site 4 consisted of available agricultural land in Pidgeon Industrial Park where sludge would be applied at lower rates as fertilizer but apparently never extended beyond a nearby 500-acre site leased by a farmer. In September 1987, TDEC transferred all regulatory authority for municipal sludge disposal facilities from SWM to WPC in accordance with the Federal Water Quality Act of 1987 amendments. New NPDES permits effective 1-29-1993 incorporated annual and monthly reporting requirements for sludge management practices under Other Requirements.

The Maxson Facility added three primary clarifiers that became functional in 1982, resulting in the production of primary sludge. Initially, the primary sludge was sent to Lagoon #1. Sludge handling facilities for primary sludge were constructed and started up during the summer of 1983. These facilities included three belt filter presses with a total capacity of 100 dry tons per day and mixing with lime for stabilization. The dewatered primary sludge was then land applied to Site 3 beginning about 1984. This practice continued until April 2004 when all primary sludge began being wasted to the lagoons for co-digestion with waste activated sludge.

During his last term in office, Mayor Dick Hackett on 8-23-1989 announced the creation of the EARTH Complex (Environmental and Resource Technology Complex) in the vicinity of the lagoons combining recycling with construction of a new landfill. An advisory board appointed

by the Mayor would plan the development of the 4500 acres of land including a land fill (1250 acres) and a wildlife preserve (770 acres). One of the lagoons would be covered to recover methane. A wood mulching/composting facility was subsequently created as well as a sod farm and pumpkin patch visited annually by elementary school children. Long term plans included the construction of a management office and auditorium. While individual elements of the plan were eventually implemented either at the site or elsewhere, the overall concept was never implemented following the election of a new mayor in 1992, the promulgation of the EPA 503 regulations in 1994, and the focus on main plant expansion to accommodate industrial growth.

Digested sludge disposal continued through 1993 by spreading, injection, or spraying liquid 1-2% sludge onto the fields by tank truck. New federal regulations known as 40 CFR 503 became effective 2-19-1994 which included Surface Disposal of sludge that met appropriate vector attraction reduction and metals criteria. The City opted to comply with federal Surface Disposal Site requirements that included containment of surface run-off and ground water monitoring and so were no longer regulated by the state program. Berms and ditches were constructed to direct surface run-off and supernatant from the approximately 600 acres of dedicated land into the Nonconnah Interceptor via a gated 18-inch line leading into a manhole on the 84-inch Horn Lake Interceptor. Lagoon #1 was covered 1995 and the flaring of biogas began. Secondary Dewatering was placed in operation in 1996 based on an umbilical system patterned after the Stiles Facility except that the two lagoons providing feed sludge were not covered. Dewatered sludge was surface disposed in the new dedicated site. The use of the sludge spreading equipment was gradually phased out and replaced with dump trucks and dozers.

An anaerobic digestion study was done in 1998 to optimize current lagoon operations, including the direct wasting of primary sludge into the lagoons. Design of new covered Lagoons 2A, 2B, and 2C soon followed and constructed in 2001-2003. Plans were made to cover Lagoon #5 later. Sale of biogas to TVA began 2000 based on a 20-year agreement. The new covered lagoons were placed in service in spring of 2004 and the plan to waste primary sludge to the lagoons was enacted, ushering in a 12-year period of double biogas production. TVA did not properly maintain their equipment and were unable to utilize a great deal of the biogas the Maxson Facility produced. Lagoon #1 cover was replaced in 2011 due to cover failures dating back to 2007. Figure 5 shows long-term accumulation of Maxson Facility surfaced disposed biosolids and sludge accumulation within the lagoons. As shown, the new Secondary Dewatering practices implemented in the mid-1990s held down lagoon sludge accumulation until 2004 when primary sludge dewatering was suspended.

The sludge accumulation rate began increasing in 2004 when primary sludge dewatering was suspended and wasting of primary sludge to the covered lagoons for co-digestion with activated sludge began. Projects were implemented to increase sludge dewatering including the refurbishment of belt filter presses (2005), the purchase of three new additional belt filter presses (2013), the replacement of the two lagoon dredges with new units (2013), and the purchase of a third dredge (2019) to mine sludge from further out in Lagoon #5. The replacement of the belt filter presses is expected in 2024. Hopefully, the current umbilical system has the capacity to keep up with the increased sludge production due to the co-digestion of primary sludge with waste activated sludge and an expansion of Secondary Dewatering will not be necessary.

Hydrogen Peroxide Application Pilot Testing in Lagoon No. 5

In 2020, the City contracted with USP Technologies (USP) to provide odor control at the TE Maxson WWTF (TEM) sludge lagoons using hydrogen peroxide (50%) dosing of digested sludge pumped to lagoon #5, which would reduce the emission of odorous sulfur compounds, especially hydrogen sulfide. USP designed and installed a hydrogen peroxide above-ground storage tank and dosing equipment (i.e., automatic dosing system). USP also provided on-going O&M and monitoring, hydrogen peroxide, periodic status reports and a web-based project dashboard.

The automatic dosing system uses a metering pump to continuously inject hydrogen peroxide into the digested sludge stream anytime the sludge pump is turned on. The metering pump is controlled by a sludge pump relay to turn both pumps on and off in concert.

As control points, USP periodically monitors dissolved hydrogen sulfide in the hydrogen peroxide treated digested sludge and gas-phase hydrogen sulfide at the digested sludge discharge point and a nearby fence-line to measure the performance of the odor control system. The hydrogen sulfide target levels at the control points are <5 mg/l of dissolved hydrogen sulfide in the treated digested sludge, <10 ppm of hydrogen sulfide vapor at the sludge discharge point, and <1 ppm at the nearby fence-line.

In calendar year 2023 a total of 63,344 gallons of 50% hydrogen peroxide was used for odor control at the TEM covered sludge lagoons at a total cost of $299,617.12. The discharge sulfides slightly eclipsed the target during Q2. There were no high peaks but dose rate was increased to combat higher seasonal sulfides. At the nearby fence-line the average hydrogen sulfide was 1.2 ppm and peaked at 13 ppm.

Interceptor Sampling Review

Tioga began collecting interceptor samples in November 2022 at weekly intervals to assess loading on each interceptor for comparison with influent loading. The sample locations also allowed for insights regarding pH and Metals. The Nonconnah Siphon downstream manhole (NO 020114) would include Blues City Brewing, Bioenergy, Kelloggs, Richardson Oil Seed, and Solae. This manhole would not include AB Mauri, Riviana, Valley, and Valero. These industries would show up in MH9.

BOD ppd	Sampled Maxson Facility Loading	Nonconnah Interceptor MH9	Horn Lake Interceptor	Nonconnnah Interceptor NOI-S	Calculated Maxson Facility Loading
Nov-22	413,506	437,540	13,341	NA	NA
Dec-22	355,484	304,093	22,435	NA	NA
Jan-23	557,339	533,780	20,383	NA	554,163
Feb-23	566,653	390,677	56,118	NA	446,795
Mar-23	430,844	456,776	13,005	NA	469,781
Apr-23	559,984	536,234	16,733	NA	552,967
May-23	312,875	392,657	20,880	NA	413,537

Jun-23	266,813	NA	14,938	NA	NA
Jul-23	334,167	NA	14,857	310,523	389,605
Aug-23	355,683	NA	16,135	231,955	312,674
Sep-23	312,325	NA	20,837	211,837	297,834
Oct-23	318,988	NA	11,392	242,700	320,706
Nov-23	343,950	339,181	9,836	203,720	282,318
Dec-23	326,236	NA	8,833	293,802	390,036
Average	389,632	423,867	18,552	249,090	402,765
Percentage	100%	109%	5%	64%	103%

The calculated loading was based on a combination of Nonconnah loading (MH9 or downstream siphon), Horn Lake Interceptor, and industrial user loading (when not already included in interceptor loading). Compared to Sampled Influent BOD Loading, on average, MH9 loading was 109%, Horn Lake was 5%, downstream siphon was 64%, and Calculated Influent Loading was 103%. No recycle/return stream impact was assumed. Sampled Influent BOD loading appears to be representative of actual loading conditions in this data set.

TSS ppd	Sampled Maxson Facility Loading	Nonconnah Interceptor MH9	Horn Lake Interceptor	Nonconnnah Interceptor NOI-S	Calculated Maxson Facility Loading
Nov-22	453963	246374	10687	NA	NA
Dec-22	509157	177142	27188	NA	NA
Jan-23	595,301	272,798	61,583	NA	334,381
Feb-23	561,582	225,916	79,971	NA	305,887
Mar-23	417,534	287,391	20,356	NA	307,747
Apr-23	436,040	273,424	11,029	NA	284,453
May-23	229,950	171,520	6,867	NA	178,387
Jun-23	198,559	NA	11,652	NA	NA
Jul-23	226,981	NA	4,053	187,738	223,889
Aug-23	197,205	NA	9,045	131,366	182,207
Sep-23	231,035	NA	36,537	117,484	192,237
Oct-23	246,597	NA	13,068	152,689	203,023
Nov-23	305,077	208,134	7,954	152,644	198,247
Dec-23	249,966	NA	10,087	197,739	252,328
average	347,068	232,837	22,148	156,610	242,071
Percentage	100%	67%	6%	45%	70%

The calculated loading was based on a combination of Nonconnah loading (MH9 or downstream siphon), Horn Lake Interceptor, and industrial user loading (when not already included in interceptor loading). Compared to Sampled Influent TSS Loading, on average, MH9 loading was 67%, Horn Lake was 6%, downstream siphon was 45%, and Calculated Influent Loading was 70%. No recycle/return stream impact was assumed. Sampled Influent TSS loading

appears to be over-reported and/or impacted by return/recycle streams not included in this data set.

		pH		
Location	Date Range	Median	Max	Min
Nonc Siphon	7/6/2023 to 12/14/23	6.1	6.7	4.5
MH9	11/7/22 to 12/21/23	5.5	10.2	4.9
HLI	11/7/22 to 12/14/23	6.7	7.5	6
PI	7/11/23 to 12/12/23	6.4	7.1	6.1
Inf	7/11/23 to 12/12/23	6.05	6.3	5.8

The above data shows the pH entering upstream of the siphon is in all cases acidic. It also shows highly variable pH at Manhole 9. The median pH at all locations varies 5.5 to 6.7.

		Molybdenum mg/l		
Location	Date Range	Avg	Max	Min
Nonc Siphon	7/6/2023 to 12/14/23	0.004	0.009	0
MH9	11/7/22 to 12/21/23	0.023	0.056	0.008
HLI	11/7/22 to 12/14/23	0	0.006	0
PI	7/11/23 to 12/12/23	0	0	0
Inf	7/11/23 to 12/12/23	0.019	0.039	0

Based just on concentration, the above data shows that less than 20% of the molybdenum originates from upstream of the Nonconnah Siphon and that very little, if any, comes from the Horn Lake or Presidents Island Interceptors. Therefore, Blues City Brewing, Bioenergy, Kelloggs, Richardson Oil Seed, and Solae would be excluded as potential major sources of molybdenum. However, AB Mauri, Riviana, Valley, and Valero, among others, are potential major sources of molybdenum.

Impact of Solae on Maxson Facility Review

The Maxson Facility had no violations in the month of June as a result of Solae being off line for at least 19 days because of a strike. The June 2023 data was compared with June 2021.

Parameter	Jun-21	Jun-23
Solae Eff BOD ppd	258,960	68,405
Solae Eff TSS ppd	185,957	42,958
Other Industries Eff BOD ppd	132,307	91,922
Other Industries Eff TSS ppd	40,252	25,635
Expected Mx Inf BOD ppd	491,267	260,327

Expected Mx Inf TSS ppd	326,209	168,593
Maxson Inf BOD ppd	611,000	264,738
Maxsn Inf BOD mg/l	1,080	516
Maxson Inf flow mdg	68	62
Maxson Inf TSS ppd	455,069	198,871
Maxson Inf TSS mg/l	802	384
Maxson Eff BOD mg/l	27	24
Maxson EFF TSS mg/l	32	28

The Maxson Facility sampled influent BOD loading and to a lessor extent the TSS loading was closer to the expected loading in June 2023 compared to June 2021. The above summary shows that the impact of Solae on the Maxson Facility influent loading was overstated in June 2021 but fairly accurate in June 2023. When controlling for this overstated influent loading, focusing solely on effluent, the ability of the Maxson Facility to process the Solae effluent discharge when they are at full production was shown to be 99% removal of all Solae effluent BOD and a 95% removal of all effluent TSS. The influent loading was shown to be overstated in June 2021 by 24% for BOD and 40% for TSS whereas in 2023, it was shown to be 2% of BOD and 18% of TSS, a vast improvement.

Disinfection Operation Overview

Continuous PAA dosing continued during 2023 at the Maxson Facility at a cost of $6,691,536 or $18,333 per day. This equates to 28 cents per thousand gallons treated at an average dose of 11.9 mg/l based on influent flow. The average pre-final effluent COD was 366 mg/l based on City lab analysis of 24-hour composite samples and 387 mg/l based on PXC instrumentation daily average.

All things being equal, the percentage of the Maxson Facility O&M expenditures spent on disinfection should be a good metric whether the operation is being properly monitored and costs contained. Over time, as experienced is gained and the process optimized, the percentage theoretically should be decreasing. Conversely, if increasing disinfection costs are resulting in reduced allocation for non-disinfection expenditures, this could be indicated. Annual disinfection costs along with total expenditures since the first year of disinfection are presented below:

Year	Total Expenditures	Disinfection Costs	Percentage
2021	$20,331,743	$2,317,939*	11.4%
2022	$30,035,390	$8,781,819	29.2%
2023	$24,746,751	$6,691,536	27%

*only includes Aug-Dec 2021 (5 months)

2.0 Stiles Wastewater Treatment Facility

Introduction-

The Stiles Facility began thawing out after the deep freeze in December. Dewatering never fully recovered and limped along for the remainder of the year. The 120-inch bypass continued with the Stiles Facility having to increase raw sewage pumping to keep the sewer level down as the contractor removed and replaced damaged pipe inserts. Heavy rains and flows up to 193 mgd on 1-3-2023 resulted in the Xylem equipment going under water. The Stiles Facility initiated a program to witness Evonik cleaning their color sensors and grab sample collection for analysis by Stiles Facility lab once per week based on the 11-28 through 12-29-2022 data set. Based on that review, Evonik overstated color 23% compared to City lab analysis comparing grab samples and by 35% comparing 24-hour samples. There appeared to have been a deterioration in Evonik instrumentation color compared to City lab color since late July 2019. Dorothy Hunt retired after 45 years, the last of the original employees at the Stiles Facility. The Stiles Facility began PAA and E coli profile sampling on 1-19-2023 to better define disinfection dynamics. Erratic PAA dosing at lower levels continued throughout the month which Evonik attributed to wear and tear of PAA feed pump (peristaltic). The system was not designed to dose below 5 mg/l indicating the company never expected to dose at the lower levels associated cold temperatures. A gravity feed system that would both provide lower feed rate and have less wear and tear was proposed. The lab unable to produce timely BOD data due to quality control issues. KTG announced that it would be shutting down a paper machine (PM1) to be upgraded over the next 6-12 months and then would start on PM3.

The lab indicated the entire BOD data for both plants for January would be flagged and COD data would be used to provide estimated results. On 2-7-2023, it was determined that BODs run on ferrous chloride nutrient solution showed 1.05 mg/l D.O. depletion which was the cause of the problem. In the meanwhile, samples were sent to Waypoint until everything could be straightened out. This ferrous chloride lot had been put into service about 1-4-23 or 1-6-23. On 2-9-2023, Gary Clancy emailed about fibers clogging up color analyzer since noon and they were operating the disinfection system in manual with a target at residual of 1 mg/l. Dosing as high as 33 mg/l occurred. It was suspected the fibers were a slug discharge related to Memphis Cellulose stopping digestion. Evonik began PAA transfers using pipeline on 2-7-2023. On 2-11-2023, American Yeast had an explosion when caustic tank blew up. Waste was stored in frac tanks and appeared to have been disposed off-site.

More slug discharges of fibers impacted disinfection from about 2-17-2023 thru 2-21-2023 corresponding to a problem with solids recovery at Memphis Cellulose reported by Industrial Monitoring and Pretreatment during 2-18-2023 thru 2-22-2023. The problem possibly started as early at 2-9-2024 based on fibers reported by Evonik so likely was intermittent from around 2-9-2023 thru 2-17-2023. During this period, Tioga reported that fibers at Memphis Cellulose always clog the pH probe with some fibers 1 foot plus in length. It is obvious that the Stiles Facility is unable to remove a significant portion of these fibers. EPA provided a draft AOC for the Stiles Facility focusing on 2-year fixes for AOCs with Consent Decrees required for longer term. Weirs began arriving for secondary clarifier weir replacement. Contractor installed new laterals for American yeast and Lagoon as part of the 120-inch rehab project with an expected completion date for the entire project by the end of April.

On 3-10-23, Gary Cohen sent a position paper regarding Stiles AOC including rehabbing 25-30 manholes in the flood plain. The Mississippi River crested at about 28 feet on 3-13-2023. Effluent Temperature was shown to be better correlated with PAA Demand when both American Yeast and Memphis Cellulose are online, meaning temperature may play an even bigger role with PAA demand during those situations. This probably has to do with the initial reaction (i.e. first ¼ tank). The greater the chemical demand, therefore, the greater the temperature impact, implying a second order reaction.

Contractor continued working on repairing various laterals related to 120-inch rehab. The Stiles drive road crossing used for bypass pumping was removed on 3-15-2023. Fish were observed on 3-30-2023 getting into the Stiles Facility influent when river stage was at 24 feet. The river stage had peaked in mid-March at about 28 feet indicating the hole opened as the river receded. The contractor began pouring tops back in launch shafts for 120-inch rehab project. In the meanwhile, CES, subcontractor to Chris-Hill, began setting up equipment and will start working on RAS Channel in April. CES previously cleaned the Front Street Interceptor and/or the 120-inch.

Color spikes at disinfection continued into April since problem at Memphis Cellulose in February occurred. As a result, Stiles Facility personnel began tracking color and PAA dosing throughout the day due to the continuing color spikes/fibers recording the data by hand from Ovation starting 4-27-2023. Stiles Facility personnel were informed on 4-18-2023 that Evonik corporate approved the conversion to gravity PAA feed project. On 4-24-2023, a test run was performed taking Stiles Facility biosolids to the Maxson Facility. The recommendation was made to increase the width of fingers so that trailer dumps could be used for dumping biosolids for back dragging.

On 4-29-2023, there was a meeting with City Information Systems about taking over the Stiles Facility EDS as a requirement for connecting Ovation to the internet so data could be accessed for the dashboard project. In addition, on 4-29-2023, Stiles Facility had full main power shut down from 5AM to 9AM to connect the new permanent emergency power generator to the Influent Pump Bldg. The two emergency power backup generators were used to power the controls and magnetic units for the first time.

Stiles Facility took a sample of fibers to Memphis Cellulose for microscopic identification 5-2-2023 but was inconclusive. Periodic fiber intrusion had been occurring since earlier in the year apparently related to problems with fiber recovery system. There was a fire at KTG on 5-2-2023 which possibly was a source of fibers. Information Systems visited the Stiles Facility datacenter on 5-2-23 in preparation to move the EDS to the Main Library later scheduled for 6-14-2023.

Rich Foods samples sent to Dr. Lane for characterization on 5-8-2023. Letters were sent to American Yeast and Memphis Cellulose on 5-17-2023 to start monitoring effluent color the beginning of July. In conjunction with the industrial monitoring, the Stiles Facility would start daily testing of influent and supernatant for apparent color as well. The monthly meeting with Tioga on 5-18-2023 revealed they were checking Memphis Cellulose pH probe weekly and could barely get the pH probe out of the holder due to fibers wrapping around it. According to Industrial Monitoring, Memphis Cellulose had installed a TSS meter to check for solids. In addition, a “breakthrough bucket” had been installed to monitor the accumulation of “mop heads.” Memphis Cellulose had to install a large mesh to filter atypical solids going to the

effluent composite sampler. If the fiber problem continues, the next steps include installing a wire mesh as both MC and at Stiles disinfection to quantity the fibers being released (i.e. pounds per hour). The monthly meetings continued with Tioga regarding carbo::lyser probes from s::can and uploading data to Digi and then to Power BI. If the system can ever work, the parameters will include BOD, COD, TOC, DOC, UV254, TSS/TS, Turbidity, and temperature. They will require regular calibration versus lab data.

The contractor continued backfilling and levelling ground after completion of the 120-inch rehab project. The Influent 480V backup generator was completed and now moving on to P1 transformer replacement at Grit Tanks. One clarifier was taken out of service and weirs removed but the replacement weirs were not fitting right. CES continued pulling debris from RAS Channel. Malasri was unable to inspect the wet well due to elevated hydrogen sulfide. Could the increase be related to completion of the 120-inch rehab? Scum continuing to cover the surface of the disinfection basin identifying the need to eventually remove the upper baffle panels.

In June, it was becoming apparent that Stiles biosolids production was 50% or lower than expected. The belt filter presses were never able to recover after the drop experienced in November 2022. On 6-16-2023, PMC Biogenics announced expansion plans. On 6-20-2023, a Mustang rental car was stolen from the admin parking lot and recovered in Millington using GPS tracker. CES continued to remove material from RAS channel with only rags being removed and not much grit, if any. Chris Hill continued working on levelling new weirs with new replacement weirs needed to feel in a low spot since the original weir dimensions were 1-1/4 inch too shallow at this location.

The month of July was characterized by aeration blower failure. On 7-5-2023, one aeration blower went down during a summer storm and parts ordered. On 7-18-2023, two blowers went down during a second freak summer storm. On 7-19-2023, there was a City wide voltage drop that caused one blower to go off-line. As a result, there was only one functional blower and TDEC was notified on 7-20-2023. The 2nd blower was back online on 7-21-2023.

On 7-10-2023, hauled waste returned to the MCS after about 26 months at TEM. On 7-12-2023, Bill Schmittou observed that the present era of foaming at the Stiles Facility, over the past 10 years, had more to do with build-up of foam in the clarifier raceways and over the disinfection tank than the aeration basins. On 7-18-2023, Evonik approved gravity feed for the Stiles Facility disinfection. Evonik would use the same system as the Maxson Facility, keeping the existing pumps upstream of existing mass meter. The discharge valve would open 4-99% up to 300 gph, requiring at least 11.1 feet of PAA for gravity to work. 96% of the flow is < 250 gph so gravity should be sufficient for most cases provided that maximum draw down of 45% instead of previous 20% is the standard for refill. PAA solution is 9.67 SG compared to water which is 8.34 SG.

On 7-19-2023, fibers were observed in prefinal effluent sample. The fibers are light enough that they do not evenly mix and appear to neither sink or float. Stiles Facility began taking pre-final effluent photos on 8-hour grabs on 7-26-2023 to document the frequently changing pre-disinfection conditions. On 7-26-2023, American Yeast had a molasses spill that went both to the sewer and on the ground beyond their plant. This created a pass thru violation for the Stiles Facility on 7-28-2023.

Andritz began rebuilding BFP #1 frame due to the belt not tracking. They had previously replaced bearings and rollers. It was learned that KTG would soon be replacing paper machine.

In August, Evonik shortened stilling wills on effluent flow meters to reduce interference from debris at the beginning of the month. Pat Jorgensen reported that the ventilation systems for the Stiles Facility influent and blower buildings originally had electric coil heating system to warm the forced air during winter but these systems had been removed years ago. American Yeast appeared to have been slug discharging molasses related to the spill intermittently from 7/31 through at 8/4/2023 before being asked to cut back. The total amount consisted of approximately 30 vactor trucks removed from the ground. The spill was caused by overloading a molasses storage tank from a barge being off-loaded and no one watching it. Stiles Facility personnel reported the Stiles Facility final sludge channels and secondary clarifier raceways to be loaded with foam. In the past, only the aeration basins were subject to having foam. It is noted that xanthum gum (used to stabilize emulsions) and guar gum (used as an emulsifier) is commonly used in the food industry. On 8-13-2023, AIC 412 spiked 2500+ color units for 1.5 hours. This appeared to be caused by high fiber content per pre-final effluent photos.

On 8-18-2023, it was learned that that EPA would give update on Stiles Facility enforcement next week.

Close observation of Ovation showed that Evonik made a 400+ point increase in color off-set on 8-20-2023 followed by a 400+ decrease on 8-22-2023. Dosing observed to be 1 mg/l of PAA per 100 color units. In person meeting with Evonik on 8-23-2023 with Greg Conrad leaving to be replaced by Andrea Johnson. At the meeting, Evonik personnel admitted making 500 +/- color unit color eternal adjustments as part of routine disinfection. This appeared to be based both on handheld meter and "gut instinct" based on perceived poor water quality. As a result of both increased Ovation monitoring and increasing familiarity with patterns related to Evonik color boosting, MCS personnel began noticing around 8-25-2023 a regular increase in color beginning about 5 pm before peaking some time before midnight during weekdays. Based on the pattern, the discovery appeared not to be related to color boosting but to an industrial discharge.

On 8-25-2023, the MCS Ovation system was finally observed at the EDS that had been relocated to the Main Library. On 8-25-2023, a 500+ point increase in color with corresponding PAA dose increase from 19 to 29 mg/l was observed at 1 pm. Evonik appears to be adjusting the color based on expected PAA residuals, becoming a de facto feed-back loop. Originally, PXC stated the disinfection tank was too small to use PAA residual for feed back control. Now, 4+ years later, they have lapsed into doing nearly everything either directly or indirectly based on PAA residuals. Not sure if the color and PAA residuals are strictly instrumental or based on lab testing. Later that day, the color increased at 5:30 pm to 1400-2000 color unit range. On 8-27-2023, it was noted that both AIC402 and AIC412 color meters could be 1000+ color units apart. This appeared to be the result of external color adjustments only being made to the color meter being used to drive the dosing with the backup meter not even close.

On 8-28-2023, it was relayed that the EPA would not pursue enforcement action at the Stiles Facility since it had been in compliance for 2 years but the decision could be revisited during a high river event.

Stiles Facility shut down influent pumping for part of 8-29 and 8-30-2023 for divers working for Chris Hill Construction and #3 sluice gate was repaired (replaced stem). Chris Hill finished coating the last manhole related to Lagoon 3N outfall replacement. Chris Hill continued installing the new weir sections where they would fit while new sections which were 2-inches shallower were being fabricated as well as a new scum hopper. The City continued waiting on replacement fans for HVAC for new maintenance building project which had been CIP. MP2 CMMS data was migrated to Hexagon in the cloud and URL updated. Past data on the MP2 computer would require too much cleanup to be migrated to Hexagon and is archived on the MP2 computer.

On 9-5-2023, based on review of Ovation data, it appeared that Evonik was making external color adjustments based on PAA residual increase which would, in fact, be subjective feedback control. While the relationship between color and dosing is objective used as a dosing parameter, boosting the color in a subjective manner using residual as a guideline would be considered feedback control. Evonik had stated their color adjustments based on handheld readings. It may be likely that Evonik was also making Base Dose adjustments based on expected/desired PAA residuals as well. The color setting should be the same as the original algorithm, possibly going back to 2019, the last full year when Evonik color closely matched City Lab color. According to the manufacturer, the Chem Scan UV 3250 external adjustment needs to be done based on laboratory grade meters in the expected mid-range of color (700-1000 color units).

Review of City influent flow compared to Evonik effluent flow after calibration showed that Evonik flow started getting lower about 1-month after calibration. If the "zero" elevation increases, then the meter will start underreporting flow. The Base Dose theoretically would need to be used to compensate for PAA dosing decreases (unless the color has been increased to compensate). Zeroing the unit should correct for any drift caused by internal issues as well as fouling. Regularly cleaning the weirs, calibrating the meters, and zero-ing the meter should address these issues.

Pre-final effluent photos from 9-11-2023 showed a progressively murky and fiber-laden pre-disinfection effluent throughout the day. Pre-final effluent photos from 9-12-2023 showed lots of solids/floating debris. The 9-13-2023 pre-final effluent photos showing debris associated with end of a Memphis Cellulose run. On 9-14-2023, both AIC402 and AIC412 dropped 300 color units mysteriously after 3 hours of apparent overdosing. Evonik was contacted on 9-17-2023 reporting that dosing was not varying with color and in the 20-25 mg/l range. Dosing was then observed to drop to 8 mg/l and trending with color in the 450-500 color unit range. There was also evidence of color spikes related to Memphis Cellulose stopping digestion due to increase in debris.

Rich Products task force met on 9-13-2023 to review data needs regarding the potential impact of this industry on MSC operations via On-Site Environmental. The types of loads received from Rich Foods need to be reviewed including any off-spec batches delivered to On-site for treatment. The 5-31-2023 samples were lowly concentrated and appeared to be rinsate. More analysis/testing of MCS race way foam is also needed. On 9-17-2023, lots of foam which was very greasy entered MCS reaching disinfection on 7pm on a Sunday.

On 9-19-2023, Evonik agreed to check/verify with Chem Scan to see if color unit had to be kept in an A/C environment to control for increase in color during the summer. Chem Scan later respond that they have units in Florida that show no temp impact. Evonik agreed that Base Dose has been increased to compensate for lower flow. On 9-20-2023, fibers inundated pre-disinfection from 630-1130 pm. All BFP were rendered out of service due to storm on 9-20-2023, taking out all VFDs. On 9-22-2023, Evonik was notified that over the past 30 days that Stiles Lab color has averaged 586 versus Chem Scan averaging 838. The handheld units, if used for external adjustment, apparently appeared to be reading 200-300 units higher than the Lab/Hach unit.

Memphis Cellulose was down for extended maintenance from 9-25 to 10-8-2023 for maintenance. The question of a hydro blasting schedule came to mind and the cleanout and discharge of tanks and heels. Additional specific information would allow a detailed evaluation of the impact of specific materials and activities on disinfection including common place slug discharges. On 9-27-2023, after effluent flow meters were zeroed and PAA residuals began increasing indicating they could reduce the Base Dose.

Planning began to take down one side of plant to clean return sludge and final channels since contractor was unable to make a dent in cleaning out the RAS channel. P1 Transformer completion date expected to be 11-3-2023 as the Influent Controls Backup Generator had been commissioned.

On 10-2-2023, an aeration blower went out on hot bearing temperature as the plant had operating two blowers at 50% indicating this mode of blower operation could increase wear and tear on the equipment. On 10-9-2023, Kazmier Associates visited MCS to evaluate effluent flow monitoring using their strap-on system. Another greasy foam discharge apparently entered MCS on 10-9-2023 going to disinfection which corresponded to about when Memphis Cellulose was to resume digestion.

The date 10-17-2023 was the lowest Mississippi River stage on record at -12.04 feet. In 2022, it was -10.81 feet on 10-21-2022, the second lowest record. The next lowest record was -10.7 feet on 7-10-1988. Tioga continued maintaining the Digi units. Eventually, the industrial monitoring pH probe will need to be moved from the effluent to the grit tanks. Stantec still unable to access Power BI where all the data is stored, effectively locked out with no idea how Brown & Caldwell set it up. MCS switched to Side A disinfection on 10-18-2023 to allow Evonik to install gravity PAA feed system. Another high color discharge at pre-disinfection on 10-25-2023 starting at 730pm, peaking at 1230 am, and ending at 830 am on 10-26-2023 where it started, starting approximately 60 hours after Memphis Cellulose stopped digestion. On 10-30-2023, the City Council punted on Sewer Ordinance Change to stop sewer extensions into unincorporated areas. They still have the option to put it back on the agenda.

The punch list was developed for the P1 Transformer Replacement Project at Grit Tanks with full completion expected in December. Still no word on the completion of the new Maintenance Building Project as the HVAC portion has become a hand full. A quote was obtained for a new Influent Wet Well Ventilation System in time to submit as part of FY24 Budget.

Evonik installed components on 11-1-2023 to set up gravity feed on Side B as MCS switched disinfection to Side A to facility this effort. MCS switched to B-Side disinfection to allow Evonik to set up the PAA gravity feed system on A-side on about 11-2-2023. The City took that as an opportunity to compare Evonik A-side and B-side effluent flows with City Influent Flow as shown below:

Scenario	Period	Stiles	Evonik	Difference
Both Sides	10/8 thru 10/17/2023	68	60	8
A side only (north)	10/18 thru 11/1/2023	69	65	4
B side only (south)	11/2 thru 11/8/2023	74	63	11
Both Sides	11/9 thru 11/10/2023	82	74	8

The difference between Side A and Side B only compared to MCS Influent combined averaged about 8 mgd compared to MCS influent flow indicating that the flow measurement problem appears to be mainly related to Side B as shown above.

On 11-2-2023, it was learned that Runyon Industries operated M-F schedule on day shift only which explains the increase in pre-disinfection color in late evening, especially apparent when Memphis Cellulose not digesting. It is likely that Evonik personnel over the years increased and decreased color dosing in response to this unknown. Three belt filter presses were running at MCS as of 11-6-2023 as the dewatering system continued to limp along. Color spikes started after Memphis Cellulose started digestion during the early part of the month. On 11-8-2023 Tioga submitted RAS samples to Marquett Univsersity for BMP analysis. Samples of RAS and On-site effluent were also submitted to Dr. Lang for foam study related to anaerobic digestion study. Samples of foam collected from the raceways will need to be submitted for surfactants and emulsifiers.

With Malasri as subconsultant to Barge, it is hoped to have the final design Stiles Biosolids Upgrade project will be completed by mid-2024. Malasri continued to wait on initiation and completion of the Influent Building ventilation project so that the MCS Influent Wet Well assessment could be completed.

On 11-9-2023, color adjustments were apparent at 530 am starting with 300+ decrease followed by a later 500+ increase. In addition, a color spike was observed after Memphis Cellulose stopped digestion indicating fiber or color-related slug discharge. On 11-13-2023, the Emergency Generator Trailer was used to run lab on 11-11-2023 during circuit repair. This was the first time the emergency generator trailer was used to power the Admin Building. On 11-14-2023, increasing pre-disinfection color trend was noticed beginning 6 pm, peaking at midnight, and subsiding by 830 am. This pattern is clearly discernable when Memphis Cellulose not digesting but can still be observed when they are digesting. This was the typical pattern associated with Runyon Industries based on the distance from the Stiles Facility and duration of discharge. On 11-15-2023, more suspected color changes related to Runyon Industries were observed, beginning at 7 pm, peaking at 1130 pm, and subsiding at 430 am, over approximately 9.5 hours, likely mirroring the period of their day shift discharge. On 11-18-2023, there was a possible color adjustment of AIC412 at 440 am from 1120 to 1647 of 500+ color units. The PAA dosing spiked from 21 to 25 mg/l at 1040 am unrelated to color. In addition, nine color

spikes were observed from 11-18-2023 thru 11-20-2023, appears to be a fiber pattern every 8-9 hours. More scum appearing on disinfection tank on 11-21-2023.

As a result, on 11-21-2023, weekly color comparison with Evonik as well as the City witnessing pre-filter cleaning was proposed. This was to insure the Chem Scan color meter pre-filters were cleaned at least weekly to reduce fiber interference/control dosing bias related to fiber intrusion and that proper color levels are maintained (i.e. no external adjustment unless compared with a consensus of lab instrumentation). A couple of trial runs were performed with weekly color checks and City witnessing pre-filter cleaning every Thursday beginning 1-1-2024. Evonik refurbished one of the PAA feed pumps with Teflon rollers and stainless steel bearings with three more to go. Evonik also planned to purchase air blast system to keep PAA residual units clear of fibers, one blast of air per PAA probe (AIC 404 and AIC 414).

On 11-28-2023, the Stiles Facility took B-Side Disinfection out of service to clean floor and remove upper part of vertical baffles to reduce scum trapping. No day was spared a color spike during 11-22-2023 thru 11-28-2023 holiday weekend after Memphis Cellulose stopped digestion with both Runyon Industries and Memphis Cellulose tank cleaning activities suspect. It was also learned that Runyon Industries had plans for a facility on President's Island that would go into service in 2024 due to discharge limits at current location.

Scum still covered disinfection tanks indicating that additional attention needs to be paid to keeping scum pumps working for the skimmer to work properly. While the first vertical baffle remained in place for PAA mixing, ports were cut into the panels to allow scum passage. Tentative plans were made to submit raceway/disinfection tank foam/scum sample to Dr. Lang for complete characterization. Planning was made for Andritz to install rebuilt BFP #5 but cracks observed in gears once returned from the rebuild resulted in more delay as the Stiles Facility continued to fall behind on dewatering.

It appeared that a 450+ external color adjustment made on 12-1-2023 on AIC412. PAA dosing jumped from 12.9 to 21.4 mg/l as a result as old habits die hard. The Stiles Facility switched to one blower beginning about 12-1-2023 at the request of maintenance to reduce wear and tear on the other remaining blower for the remainder of the month. Elevated E coli was detected on 12-8-2023 and 12-9-2023 due to rapids build-up of foam at the back end of the plant.

Rich Foods Task Force meeting on 12-13-2023. BMP results were back from Marquette University showing 197-211 ml CH_4/ g COD at 35 deg C and 175-187 ml CH_4/g COD at 0 deg C for On-site effluent collected about 10-30-2023. The 2013 data showed WAS = 264, digested WAS = 38, and American Yeast beer waste = 183 mg CH_4/ g COD. So, from a BMP perspective, On-site discharge is comparable to Beer Waste. The 10-30-2023 On-Site effluent submitted to Dr. Lang for foamability showed that digestion increased foam creation by a tremendous amount. In fact, the entire test tube was foaming. The On-Site discharge permit is believed to be up for renewal in 2026. On 12-21-2023, the M-F evening pattern of color increase/decrease at pre-disinfection continued to be observed, especially apparent after Memphis Cellulose stopped digestion and not digesting from 12-20-2023 through 1-3-2024. On 2-27-2023 and 12-28-2023, daily maximum E coli exceedances not related to scum were observed, indicating possible process upset.

The Maintenance Department performed numerous noteworthy tasks during the year.

In January, #3 Dragout was rebuild was completed. In February, the North side wash press rebuild was completed and a new seal water sump pump was installed. In March, a new wet well 24-inch' ventilation fan was installed. In addition, permanent 8-inch and 6-inch HDPE piping for electric emergency submersible pumps were installed at the influent dry well. Permanent 8-inch HDPE piping for hydraulic emergency submersible pumps and permanent 1inch hydraulic lines were installed. Bad brick around the exterior of the Admin Building was replace. In April, Lagoon 1 and 2 walkways were replaced from the bank to the outfall structure. In May, a new heat exchanger was installed on #4 Blower. In June, the incinerators demolished and incinerator building was remodeled for future purposes.

In August, the #5 Fine Bar Screen Rake was rebuilt and a brand new gearbox for #5 Fine Bar Screen rake was installed. A leak in the 6-inch supply line to dewatering was repaired and new 6-inch valve installed. In September, #3 Raw Sewage sluice gate stem was replaced. A new shut off valve was installed for #3 blower water supply and Course Bar Screen #2 chain replaced. In October, a new effluent sample pump was installed. In November, a complete liner change was performed in North Screw Conveyor. In December, a new shut off valve was installed for #1 blower water supply and Course Bar Screen #3 chain was replaced.

In addition to the major tasks listed above, the Maintenance Department completed 3063 Preventative Maintenance Work Orders, 632 Scheduled Maintenance Work Orders, and 11 Unscheduled Maintenance Work Orders for a total of 1317 work orders. The annual number of work orders completed over the last 10 years is summarized as follows:

Year	2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
Preventative	3063	963	NA	2853	3178	2795	3148	2694	2738	2802
Scheduled	632	NA	NA	286	531	758	497	345	552	799
Unscheduled	11	354	NA	599	239	255	188	138	156	192
Total	3706	1317	NA	3912	4013	3863	3857	3177	3446	3810

Flow – The average daily influent flow for the Stiles Facility in 2023 was 85 mgd. This is a decrease of 5 mgd from the previous year. The 0.5th percentile flow in 2023 was 61 mgd which is considered base flow for the year. The average daily flow from the major significant industrial users was 13.6 mgd. This leaves 10.4 mgd or 88% of average daily flow accounted for. The rainfall was 58 inches (plant gage). The average annual rainfall at the Stiles Plant gage has been 55 inches.

BOD_5 – The average influent BOD_5 in 2023 was 288,533 pounds per day. This is 21,111 pounds or 9% more loading from the previous year. The average influent BOD_5 concentration was 414 mg/l and the average effluent concentration was 26 mg/l for a BOD_5 removal efficiency of 94%. The average effluent BOD concentration adjusted for PAA was 21 mg/l for a BOD removal efficiency of 95%. The total average BOD_5 in pounds per day reported discharged by the major significant industrial users was 144,395 ppd in 2023 which equals to 50% of the total loading. This is 2019 pounds per day or 1.4% less than the previous year. The 0.5 percentile loading in

2023 was 119,054 ppd which is estimated domestic loading for this year. This leaves 25,084 ppd of BOD or 91% accounted for.

Total Suspended Solids (TSS) – The influent total suspended solids averaged 240,561ppd in 2023. This is 7771 pounds per day or 3% more than the previous year. The average influent TSS concentration was 344 mg/l and the average effluent concentration was 11 mg/l for a TSS removal efficiency of 97%. The total average TSS reported discharged by the major significant industrial users was 57,032 ppd in 2023 which equals to 24% of the total loading. This is 2088 pounds per day or 4% more from the previous year. TSS loading from lagoon supernatant and dewatering filtrate return was sampled at 34,712 ppd or 14% of total average daily TSS loading. The 0.5 percentile loading for 2023 was 114,387 ppd which is estimated domestic loading for this year. This leaves 34,430 ppd of TSS or 86% of average daily loading ccounted for.

Chemical Oxygen Demand (COD) - The influent averaged 915 mg/l and 632,884 ppd which is 9552 ppd or 2% increase from the previous year. The effluent averaged 93 mg/l and 65,928 ppd which would include PAA. The 0.5 percentile loading for 2023 was 265,055 ppd so the estimated domestic loading for this year is 265,055 ppd. The total average COD discharged by the major significant industrial users was 302,023 ppd which equals 48% of the total loading. This leaves the remaining 65,806 ppd or 90% of influent COD loading ccounted for. The COD surcharge is based on covering 50% of disinfection costs.

Effluent E. Coli - Effluent E. analysis was performed on daily effluent grab samples throughout the year as required by the NPDES permit as report only. Effluent E. coli results for this past year are presented in Figure 6 based on disinfected effluent.

Water – In 2023 the metered water usage totaled 139,774 CCF at a cost of $184,300. The average metered water consumption was 13,977 CCF per month or 0.3 MGD. This is a decrease of 2598 CCF per month from the previous year which indicates the flows are potentially being bypassed around the meter. Flows discharged directly from American Yeast for use by the Stiles Facility (i.e. aeration basin and scum hopper spray) averaged 1.3 mgd.

Energy – The plant main power consumption was an average of 5,230,635 kwh of electricity per month in 2023. This is approximately 723,785 kwh per month less than the amount used in the previous year. The monthly peak demand averaged 8154 kw which is 639 kw less than the previous year. The average monthly main power bill in 2023 was $419,456. This is $64,167 less per month than the previous year. Payments for participating in the ENERNOC curtailment program for main power are not available but are expected to be minimal due both to inability to reduce raw sewage pumping rate due to construction and increasing difficulty and concern with restarting aeration blowers. It is believed that the co-gen fired by natural gas and serve this function in the future.

April was the high month for demand at 8462 kw and January was the high month for consumption at 5,879,409 kwh. December was the low month for demand at 7359 kw and December was the low month for consumption at 3,842,372 kwh. Hourly electric demand throughout 2023 is shown in Figure 2.

Effective 10-1-2015, the City entered into an agreement with MLGW to switch from Seasonal Demand to Time of Use billing. TVA would charge Fuel Cost Adjustment based on user class.

TOU rate structure benefits high load factor customers through incentivizing off-peak energy usage. The contract term is 5 years and will be automatically extended unless otherwise terminated. The City signed an alternate circuit agreement with MLGW on 10-2-2019 for 10 years for the circuit switches to be on the MLGW power poles. This will result in an additional $13,778.23 monthly charge on the main power bill once the program is in place.

Sludge and Biosolids – The Stiles Plant wasted 110.1 million dry pounds of secondary sludge to the lagoons in 2023. This is 2.6 million pounds more compared to the previous year. 25.8 million dry pounds was surface disposed or land applied at a 82.2% capture rate.

The annual amount of digested sludge (in million dry pounds) removed from the two covered lagoons over the last 10 years is provided below:

2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
25.8	40.1	38.3	36.1	38.8	43.9	31	44	30	29.4

Approximately 34% of the secondary sludge wasted to the lagoons in 2023 was destroyed. The formula used to calculate percent destroyed is %VS of WAS multiplied by %VAR. This is 37.4 million dry pounds (18,717 tons). 7.6 million dry pounds was removed from Lagoon 3N by the contractor and disposed at the landfill. 5.3 million dry pounds was removed from Lagoon 3S by another contractor and disposed at the landfill. Considering removal and volatilization the new change in the lagoon solids inventory was an increase of 22 million dry pounds within Lagoon Nos. 1, 2, and 3. The calculated cumulative dry solids within the three lagoons is 838 million dry pounds. In 2023, a net approximately 32 million dry pounds (16,000 dry tons) of bio-solids were removed from Surface Disposal Site and about 58.5 million dry pounds (29,250 dry tons) were land applied, landfilled or surface disposed at the Maxson Facility. The calculated dry bio-solids that has been placed in the Surface Disposal Site since 1984 is 1382 million dry pounds. This is the first recorded net decrease in the Surface Disposal Site amount.

For the last 10 years, net accumulation has been occurring at an average rate 30 million dry pounds per year in Lagoons 1, 2, and 3. Sludge removal from the three lagoons has averaged 35 million dry pounds per year. To stop net accumulation, the Stiles Facility will need to dewater and surface dispose or land apply and/or clean out lagoons and dispose an average of 65 million dry pounds per year which is 187% the existing rate of removal or destruction.

The total cost for the dewatering and surface disposal/land application/landfill disposal operation was approximately $11,248,259. This equates to a unit cost of $208 per dry ton of bio-solids surface disposed, landfilled, land applied and volatilized in 2023. The annual cost per dry ton for disposal/land application and volatilization combined over the past 10 years is as follows:

2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
$208	$232	$142	$94	$104	$83	$93	$85	$102	$70

Generally, the unit cost to dewater and dispose sludge as bio-solids is a function of polymer cost and dewatering throughput. The cost to dewater and surface dispose, landfill and land apply digested sludge was $315 per dry ton in 2023. The annual cost per dry ton based on dewatering throughput over the past 10 years is as follows:

2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
$315	$367	$233	$207	$186	$169	$250	$168	$129	$199

Biogas- Approximately 398.1 million cubic feet of biogas was produced 2023. The annual biogas production over the last 10 years is as follows (in millions of cubic feet per year):

2023	2022	2021	2020	2019*	2018*	2017*	2016*	2015*	2014*
398.1	391	396.7	349	172	167	148	151	275	427

Years marked * have been revised upward and are based on blower electrical consumption rather than flow meter readings due to faulty flow meter. The annual daily average bio-gas production over the last 10 years is as follows (in million cubic feet per day) based on revised data:

2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
1.09	1.07	1.09	0.95	0.47	0.46	0.41	0.41	0.75	1.17

The biogas produced equates to 10.8 cubic feet of biogas per pound of volatiles destroyed. The yield (ratio of biogas to pounds of volatiles destroyed) during the 10-year period based on revised 2013-2019 data is shown below:

2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
10.8	12.9	10.2	8	6.3	4.2	3.5	3.9	8.1	12.9

The amount of sludge volatilized each year appears to be constant. This could be the result of a fixed quantity of inert solids being dredged and dewatered and not associated with fresh input and biasing the amount of volatile reduction on the high side in these lagoons. It could be that a portion of the sludge is not actually being destroyed but is accumulating as an isolated volatile fraction that is remaining in the lagoon and not being dewatered. This isolated volatile fraction is not being properly mixed and broken up to allow anaerobic microbes to digest it due to short-circuiting and contributes to the reduction in viable lagoon volume. Trends in biogas production over the years are shown in Figure 7. The increase in biogas production beginning in 2021 is believed to be due to long periods of time between switching lagoons for wasting which allows for the breakup of the isolated volatile portion and increases biogas yield. Even though the quantity of amount of volatile reduction appears to be constant from year to year, the quality of the volatiles being produced (i.e. that can convert into biogas) has improved as the biogas yield has been constantly improving since 2017. This could be related to the practice beginning in late 2020 of dredging from the lagoon year-round as opposed to April-November only which was previous practice.

In 2023, American Yeast received zero million cubic feet of biogas from the Stiles Facility which is zero % of total biogas production. The revenue received from American Yeast for this biogas was zero $. The annual usage by American Yeast for the last 10 years along with the amount of biogas flared as waste gas by the Stiles Facility including revised 2014-2019 data is as follows (in millions of cubic feet):

Year	2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
American Yeast	0	91.86	128.15	152	48.9	71.9	85.3	63.1	80.5	84.3
Flared	398.1	391	269	197	133	95.1	62.7	87.9	194.5	342.7

Permit Violations- This year the Stiles Facility had two (2) NPDES Permit limit excursions related to treatment efficiency and performance.

In June, there was one permit excursion. There was a BOD daily maximum concentration excursion on 6-24-2023. This was attributed to a bad grab sample being collected due to the autosampler being out of order because of a lightning strike.

In July, there was one permit excursion. This was a BOD daily maximum concentration excursion 7-28-2023. This was due to a slug discharge of raw molasses by American Yeast.

The Stiles Facility has had 1002 permit violations for treatment efficiency and performance since the state permits were issued in 1982. The cumulative violations are shown in Figure 4 with Maxson Facility cumulative violations for comparison. The excursions associated with different major periods of plant history are as summarized below:

Period	Average Number of Excursions Per Year	Description
1982-1984	106.7	Commissioner's Order
1985-1992	6.5	Municipal Compliance Plan Annual Reporting
1993-2010	14.5	Occasional collection system failures and high river stages
2011-2023	28.5	Increasing collection system failure and high river stages.

The Stiles Facility has had 5 years (1987, 1992, 1999, 2001, and 2007) with zero violations. Excluding years of extended high river stage and/or major collection system failure, the Stiles Facility has averaged 2.2 violations per year (24 out of 41 years). The highest number on record was 2019 with 181 violations.

The pounds-based parameters were eliminated from the Stiles Facility permit beginning in 2011 with only concentration and percent removal parameters remaining for BOD and TSS. The Stiles Facility number of violations potentially could have been higher had the pounds permit parameter not been eliminated.

The violations per year on a percentile basis since 1982 are summarized below with Maxson Facility for comparison:

Percentile	Maxson	Stiles
0.25	3	2
0.5	10	5
0.6	21	8

0.625	23	12
0.65	24	12
0.7	27	21
0.75	31	31
0.8	48	41
0.85	65	52
0.9	125	88
0.94	610	125
0.95	812	146
0.96	1062	161
0.97	1338	173

As shown above, the Stiles Facility on an annual basis has a somewhat comparable level of violations compared to the Maxson Facility up to the 75th percentile level. Beyond this level, the Maxson Facility has more violations.

Bio-solids Long Term Planning

In 1971, planning was initiated by the City Division of Engineering to turn municipal garbage into liquid fuel by processing it at a facility to be located at Airways and Democrat and piping the fuel to the TVA Allen Steam Plant. TVA subsequently turned down the proposal. In 1973, when construction of the Maxson Facility started, planning was initiated to incinerate both garbage and municipal treatment plant sludge and sell the steam to TVA. In 1974, the City executed a contract with Wegman Engineering to perform a feasibility study of the construction of incinerators for this purpose. In the same year, MLGW announced plans that it would construct a garbage incinerator to provide steam for downtown buildings. In response, the City expanded its scope of incineration in April 1975 to include flammable liquids. The two competing proposals were submitted to the City Council for review and the MLGW plan was selected on 9-9-1975. In January 1976, in an apparent compromise, the City Council agreed to give half of the garbage to MLGW to generate steam for downtown office buildings and half to the City to generate steam for the Allen Steam Plant. The City was able to qualify construction of its proposed incinerators under 201 Facilities Planning and included it in the 1977 Construction Grants application submitted to the US EPA. Municipal garbage would be ground up and then trucked to the Maxson and Stiles Facilities for co-incineration with the sludge. MLGW abandoned its incinerator project in early 1978 when its consultant determined that the City had overestimated its garbage production. On 8-9-1978, EPA gave preliminary approval for the incineration project and City Council approved on 8-22-1978. However, on 10-6-1978, the EPA announced that it would only fund 33% of the cost or $49 million since that was the portion of the project related to municipal sludge and the City would have to pay the remainder or $130 million. The City had hoped the EPA would fund 75% as with other wastewater projects. In the meanwhile, sludge continued to accumulate in lagoons at both treatment plants, including Lagoons 1 and 2 at the Stiles Facility. The EPA gave the City a final deadline of 9-1-1979 to make a decision whether or not to accept the grant, indicating that potential enforcement would be taken if proper sludge disposal arrangements were not made. The deadline was passed without a decision being made.

The Stiles Facility generated 245 million dry pounds of sludge after the treatment plant first opened from 1978 to 1982. In 1980, the City designated a 53-acre area adjoining the south end of

the airport runway as a temporary bio-solids landfill known in the early planning documents as Sludge Landfill No. 1. A scope of operation plan dated 4-14-1981 was prepared for this facility and approved by the TDHE. On 4-21-1981, the Director of Public Works informed the City Council that the City had 608 days of sludge storage capacity left at the Maxson Facility and 375 days left at the Stiles Facility.

During 1980 to 1983, the plant began dipping sludge out of the lagoons and disposing it in Sludge Landfill No. 1 of which only 35 acres was used. There was no dewatering performed at that time as the original dewatering facility was improperly designed so a second contract was issued to build a another. Lagoons No. 3 and No.4 comprising 65 acres were permitted by TDHE as a sewage sludge landfill on 1-18-1984 with requirements for groundwater monitoring and runoff control. This was known as Sludge Landfill No. 2. The City constructed Lagoon No. 4 and began using it in the summer of 1984. New federal regulations known as 40 CFR 503 became effective 2-19-1994 which included Surface Disposal of sludge that met appropriate vector attraction reduction and metals criteria. The City opted to comply with federal Surface Disposal Site requirements that included containment of surface run-off and ground water monitoring which the City was currently doing and so was no longer regulated by the state program. This became established as the Surface Disposal Site for the next 25+ years.

Under Pathogen Reduction for Surface Disposal, the plant has been reporting under Class B, Process to Further Reduce Pathogens, Alternative 2 (anaerobic digestion). Under Vector Attraction Reduction, the plant has reported under Alternatives 1 and 2 (greater than 38% volatile reduction, less than 17% additional volatile reduction) based on waste activated sludge as the starting point and feed solids as the endpoint for Alternative 1. Prior to 2002, dewatered cake was the endpoint for Alternative 2. Beginning in 2002, feed solids were the endpoint for Alternative 2.

Figure 8 shows the calculated cumulative net change in material within Lagoon Nos. 1, 2, and 3 and the cumulative dry solids stored in the Surface Disposal Site. The Surface Disposal Site is at approximate full capacity and new biosolids are being diverted to land application sites pending a review of alternatives regarding the Surface Disposal Site.

Based on record keeping and volatile reduction calculations, the amount of material stored in the three lagoons is 362,000 dry tons. An additional 18-20% reduction of the original material may have occurred due to longer term volatilization given the extreme 35+ year holding period. This would put the in-situ quantity at 289,600 dry tons. Lagoon 3S is believed to be near sludge storage capacity based on an investigation completed during the months of November and early December 2016 with an average total solids content of 15%. Lagoon Nos. 1 and 2 can remain useable if the City can reduce the net accumulation rate. The City continued drying out Lagoon 3N using heavy equipment to gain use of the area as a future surface disposal site and used it on occasion for additional surface disposal due to worsening surface conditions in the Surface Disposal Site. In 2020, planning was initiated to construct a sump at the northeast corner of Lagoon 3 North to discharge into an outfall connected to the adjoining interceptor sewer. Chris Hill Construction was selected to perform this task and work initiated in late 2021. The sump and connection to the Loosahatchie Interceptor was made April 2022 with chemical coating of manholes initiated after completion of the 120-inch pipe insertion project. A proposed biosolids master plan was presented to the Director on 3-22-2023 focusing on options for the Stiles

Facility Surface Disposal Site and using the existing 870-acre Maxson Facility surface disposal site be used for both Maxson and Stiles Facility biosolids. He emphasized the City's need to utilize the adjoining DeWitt Spain Airport property for Stiles Facility biosolids management as well as PFAS impact on land application. The additional land would certainly come in handy for solar drying material generated from lagoon clean-out prior to off-site disposal, etc. In August 2023, Chris Hill finished coating the last manhole related to Lagoon 3N outfall replacement.

On 6-3-2020, a contract with Denali was executed to provide land application of biosolids on farmland and/or landfill disposal through 8-1-2023 with two additional 1-year extensions. Land application of Stiles Facility biosolids continued through 2023. On 3-31-2023, Denali quoted $36.12/WT to haul Stiles biosolids off the belt via the pit to Maxson Facility. In June 2023,10-12 loads of Stiles Facility biosolids arrived at the Maxson Facility for surface disposal on 6-13-2023. It was observed that the product was very watery and filled with maggots at least visually indicating that VAR was not being met. Intermediate planning was implemented to rebuild the presses due to uncertainty and timing of the CIP project to replace the belt filer presses with centrifuges.

A kickoff meeting for Stiles biosolids project consisting of sludge digestion and dewatering improvements was held on Teams 1-12-2021. Design will be 30%, 60%,90%, and Final. Project will include EQ Basin design which will take up all of Lagoon 3N. The project will include a new 2-story centrifuge building with adjoining sludge screening and septage receiving. A 30% design was submitted 3-29-2021 including Basis of Design Report. A 60% design was submitted on 3-29-2021. Barge submitted a draft 90% design on 5-6-2022 for review. The design included centrifuges, an Equalization Basis to be located between Lagoons 3S and 3N, a new Lagoon 3 in the vicinity of Lagoon 3S, new flares, and a sulfur scrubber to treat biogas. The draft 90% review for the Stiles biosolids project was performed on 6-15-2022 and comments provided. In September, samples of lagoon cover were collected to evaluate the possibility of leaving the existing lagoon cover perimeter flap in place for attaching a new cover. An attempt was made to schedule a pilot test with a vendor promising to remove sulfur from bio-gas. The project was submitted to TDEC SRF on 12-29-2022 for funding for both the biosolids and EQ basin. No funding approval was received in 2023.

In March 2023, Eagle Dynamics met with Stiles Facility regarding Lagoon 3S cleanout. They can provide transportation to landfill. The 3-21-2023 proposal from Eagle Dynamics to remove 9150 WT from Lagoon 3S was reviewed. This would be 2471 DT out of 100,000-110,000 dry tons or 2.4% of the total. Proposal is $2,143,141. Therefore, the cost to dredge and landfill the entire contents of 3S could be $89.3 million. Chris Hill did the Lagoon 2A project at the Maxson Facility for $791.4/DT. Eagle Dynamics proposal works out to $750/DT. In June 2023, Eagle Dynamics began setting up to remove material from Lagoon 3S. By September 2023, Eagle Dynamics continued removing material from Lagoon 3S, having removed 3.3 million dry pounds by mid-month. In October, Eagle Dynamics continued dredging Lagoon 3S having removed 483,900 dry pounds so far and getting 29-30 % cake solids. On 11-27-2023 Eagle Dynamic completed the Lagoon 3S removal project, removing about 5.3 million dry pounds from Lagoon 3S in 2023 during a period of about 5 months, leaving approximately 215 million dry pounds for the future.

Disinfection Operation Overview

Continuous PAA dosing began on 11-10-2018 at the Stiles Facility and continued through the year at cost of $15,048,244 or $41,288 per day. This equates to 48.5 cents per thousand gallons treated at a calculated average dose of 12.6 mg/l (based on influent flow). The average pre-final effluent color was 509 Pt Co based on City lab analysis of 24-hour composite samples and 858 Pt Co based on PXC instrumentation daily average (AIC 412).

All things being equal, the percentage of the Stiles Facility O&M expenditures spent on disinfection should be a good metric whether the operation is being properly monitored and costs contained. Over time, as experienced is gained and the process optimized, the percentage theoretically should be decreasing. Conversely, if increasing disinfection costs are resulting in reduced allocation for non-disinfection expenditures, this could be indicated. Annual disinfection costs along with total expenditures since the first year of disinfection are presented below:

Year	Total Expenditures	Disinfection Costs	Percentage
2019	$35,471,331*	$14,915,064**	42.1%
2020	$26,286,152	$10,626,522	40.6%
2021	$30,915,816	$12,680,787	41.0%
2022	$33,762,346	$10,975,110	32.5%
2023	$40,992,619	$15,048,244	36.7%

*includes Nov-Dec 2018 total expenditures

**includes Nov-Dec 2018 disinfection costs

Impacts on Final Effluent BOD

Continuous PAA dosing began on 11-10-2018. The actual PAA usage for 2023 and calculated dosage of active ingredient based on influent flow is provided as follows:

Month	Amount of 15.5% PAA Solution Used (pounds)	Calculated Average Dose of Active Ingredient (mg/l)	Increase in Soluble BOD Between Pre-Final and Final Effluent (mg/l)	Final Effluent BOD (mg/l)	Adjusted Final Effluent BOD (mg/l)
January	1,481,018	8	-1	29	29
February	1,546,019	10	6	27	22
March	1,768,102	8	7	25	17
April	1,782,211	10	8	28	20
May	2,153,356	14	7	23	16
June	2,708,530	18	14	38	24
July	3,033,667	19	12	34	24
August	2,423,436	15	0	22	21
September	1,883,240	14	6	28	21
October	1,481,571	12	1	20	20
November	1,527,466	12	3	20	17
December	1,462,317	11	2	25	23
Average	1,937,578	13	6	27	21

Color Contributors and Impact on Disinfection

The Color Contributors impacting the Stiles Facility are in four groups consisting of Memphis Cellulose, Runyon, American Yeast, and All Other Users. The other major differentiator in PAA dosage is whether Memphis Cellulose is online. The following statistics were derived for various parameters during 2023:

Parameter	Memphis Cellulose Down	Memphis Cellulose Running	p-value	Overall Average
PAA Solution Usage ppd*	55,868	77,959	0.053	66,399
PAA mg/l*	15.4	20.4	0.054	17.8
City PAA residual mg/l	2.0	2.0	0.304	2.0
Influent COD mg/l	870	965	0.048	915
Pre-final Effluent COD mg/l	81	107	0.002	93
Final Effluent COD mg/l	76	97	0.000	86
Influent Flow mgd	81	89	0.448	85
Effluent Flow mgd*	62	69	0.000	66
E coli (99 percentile)	2420	2420	NA	2420
E coli (geomean)	51	47	NA	48
Number of Daily E coli Permit Exceedances	12	3	NA	15
Influent Color-City	2165	2803	0.441	2426
Influent True Color-City	580	888	0.002	684
Pre-final Color-City	446	578	0.000	509
Pre-final True Color-City	167	361	0.000	230
Pre-final Color (AIC 412)*	719	1010	0.050	858
Pre-final Effluent sBOD mg/l	13.5	17	0.150	15.2
Final Effluent BOD	24	29	0.137	26
Increase in sBOD mg/l after Disinfection	4.4	6.8	0.007	5.6
Pre-final Effluent BOD mg/l	25.4	32.2	0.000	28.7
Days Down or Running	191	174	NA	NA

*Based on Evonik data from Ovation

The above statistics reveal a lot of useful information. When Memphis Cellulose is on-line, for example, influent COD concentration increases from 870 to 965 mg/l or 11%. Approximately 28% of this COD increase appears as additional COD in pre-final effluent sampling (performed upstream of disinfection). The increase in pre-final COD concentration is associated with a 30% increase in pre-final Apparent Color concentration and a 116% increase (more than double) in True Color concentration. Another observation is that when Memphis Cellulose is on-line, pre-final sBOD increases from 13.5 to 17 mg/l and final effluent BOD increases from 24 to 29 mg/l. The sBOD increase after disinfection is 4.4 mg/l when Memphis Cellulose is off-line and 6.8 mg/l when on-line. This indicates that a minimum of 2.4 mg/l of the increase in final effluent BOD when Memphis Cellulose is on-line could be attributed to PAA dosing. This is because additional PAA dosing is required when Memphis Cellulose is on-line.

In addition, the impacts of Memphis Cellulose on Pre-disinfection effluent quality can also be ascertained as shown on the following table for 2023:

	Actual Total Influent Concentration with Memphis Cellulose On Line mg/l	Prefinal Effluent Increase with Memphis Cellulose On Line mg/l	Percent Pass Through to Disinfection for Memphis Cellulose
COD	965	27	28%
Apparent Color	2803	132	20%
True Color	888	194	63%

Based on the above table, 72% removal is provided for Memphis Cellulose COD, 80% removal is provided for Apparent Color, and 37% removal is provided for True Color. The impacts of all other industrial users (apart from Memphis Cellulose) on Pre-disinfection effluent quality are shown on the following table for 2023:

	Influent Concentration Increase with Memphis Cellulose On-line mg/l	Discharge for all other users mg/l	Prefinal Effluent Discharge for all other users mg/l	Percent Pass Through to Disinfection for all other users
COD	96	869	80	9%
Apparent Color	638	2165	446	21%
Tue Color	308	580	167	29%

Based on the above table, for all other industrial users, 91% removal is provided for COD, 79% removal is provided for Apparent Color, and 71% removal is provided for True Color. Further analysis shows the percent pass through to disinfection for all users except Memphis Cellulose and Runyon as shown below:

	Influent Concentration for All Users Except Memphis Cellulose and Runyon	Prefinal Concentration for All Users Except Memphis Cellulose and Runyon	Percent Pass Through to Disinfection for all other users except Memphis Cellulose and Runyon
COD	778	75	10%
Apparent Color	2116	450	21%
True Color	477	150	31%

The impact of Memphis Cellulose on Final Effluent quality is shown on the following table for 2023:

	Memphis Cellulose Status		
Parameter	On-Line	Off-Line	P-value
Final Eff BOD mg/l (adjusted for PAA impact)	22.3	19.6	0.473
Final Eff BOD mg/l (not adjusted for PAA imact)	29.2	24	0.137
Stiles Final Eff TSS mg/l	12.3	10.1	0.01
Days	174	191	NA

As shown above, during 2023, the Stiles Facility was able to handle the out-sized impact of Memphis Cellulose on Final Effluent quality on an annual basis.

The impact of Runyon Industries on PAA dosage and impact of pre-disinfection effluent quality is readily becoming apparent. Runyon Industries is typically a day shift only, Monday through Friday operation. To discern its impact, a data evaluation was performed with and without Memphis Cellulose on-line. Note that the weekend and holiday data which is when Runyon is typically off line may be biased lower since other industrial users may have a similar schedule.

	Memphis Cellulose Not Digesting		Memphis Cellulose Digesting	
	Runyon Off Line*	Runyon On Line	Runyon Off line*	Runyon On Line
PAA Solution ppd	55,357	56,163	79,314	77,514
PAA dose mg/l	15.5	15.4	21.9	19.9
Stiles Final Eff PAA mg/l	2.0	1.9	1.7	2.1
Influent COD mg/l	778	923	887	990
Pre-final COD mg/l	75	84	93	111
Final Eff COD mg/l	74	77	89	100
Stiles Influent Flow mgd	80	82	82	91
E coli (99 percentile)	1348	2333	1600	1832
E coli (geomean)	53	48	54	45
Number of E coli Daily Permit Exceedences	4	5	1	2
Stiles Pre-final Eff Color	450	444	553	587
Pre-final AIC 412 Eff Color	693	733	1090	983
Pre-final True Color	150	174	383	349
Pre-final sBOD mg/l	10	16	11	19
Pre-final Eff BOD mg/l	23	27	25	36
Final Eff BOD mg/l	23	25	25	31
Increase in sBOD after disinfection	5.7	3.6	7.1	6.8
Final Eff Adj BOD mg/l	17	21	18	24
Days Running or Down	70	121	43	131

*Note that data with Runyon Industries off-line will be more representative of weekends and holidays.

Runyon appears to have an outsized impact on pre-final and final effluent sBOD/BOD with and without Memphis Cellulose being on-line. In addition, the impacts of Runyon on Pre-disinfection effluent quality can also be ascertained as shown on the following table for 2023:

	Influent Concentration Increase with Runyon on line* and Memphis Cellulose off line	Prefinal Concentration for All Users Except Memphis Cellulose and Runyon	Pre-final Concentration Increase with Runyon on line* and Memphis Cellulose off line	Percent Pass Through to Disinfection for Runyon*
COD	145	75	9	6%
Apparent Color	67	450	-6	0%
True Color	148	150	24	16%

*Runyon's impact is likely more than twice the value shown since it is experienced over 10-12 hours.

As shown above, the True Color contributed by Runyon potentially had greatly more impact on pre-disinfection effluent than COD. The impact of Runyon is masked by the fact that its discharge is received over a shorter duration starting late evening and not over a 24-hour period like the other industrial users and could be better described as a series of batch discharges. The impact of Runyon on final effluent quality under various scenarios is shown below:

	Memphis Cellulose Not Digesting		Memphis Cellulose Digesting	
Parameter	Runyon Off line	Runyon On line	Runyon Off line	Runyon On line
Final Eff BOD (adjusted for PAA impact)	17	21	18	24
Final Eff BOD (not adjusted for PAA impact)	23	25	25	31
Stiles Final Eff TSS conc	11	10	11	13
Days	70	121	43	131

Based on the above table, for all other industrial users, 90% removal is provided for COD, 79% removal for Apparent Color, and 69% removal for True Color. More than likely, the majority of the True Color passing through to disinfection from the Other Users is from American Yeast.

An attempt was made to estimate impact of American Yeast, the third major color contributor, on influent and pre-disinfection effluent. The last week of the year, 12-24-2023 through 12-31-2023 was used as a baseline during which time American Yeast activities are minimal. The following comparison was made to evaluate American Yeast potential impact:

The above table shows that 28% of the True Color potentially contributed by American Yeast passes through to disinfection.

	All Users Except Memphis Cellulose and Runyon	12-24 thru 12-31-2024	Estimated American Yeast values
Influent COD mg/l	778	890	0
Pre-disinfection COD mg/l	75	79	0
Influent True Color	477	287	190
Pre-disinfection Color	150	96	54
Influent Apparent Color	2116	2350	0
Pre-disinfection Apparent Color	450	602	0

The following table shows the estimated impact of the three major color contributors on influent and pre-disinfection effluent at the Stiles Facility:

	COD mg/l	True Color	Apparent Color
Influent-Memphis Cellulose	96	308	638
Pre-disinfection effluent-Memphis Cellulose	27	194	132
Pass Through-Memphis Cellulose	28%	63%	21%
Influent-Runyon*	146	148	67
Pre-disinfection Effluent-Runyon*	9	24	0
Pass Through-Runyon*	6%	16%	0
Influent-American Yeast	0	190	0
Pre-disinfection Effluent-American Yeast	0	54	0
Pass Through-American Yeast	0%	28%	0%
Influent-All Other Users	778	287	2116
Pre-disinfection Effluent-All Other Users	75	150	450
Pass Through-All Other Users	10%	52%	22%

*Runyon's impact is likely twice the value shown since it is experienced over 10-12 hours.

The 52% pass through rate for True Color for all other users is likely due to recycle streams from the lagoons as well as return activated sludge. Influent COD could be over-reported to a certain extent which could mask the contribution from American Yeast. Apparent Color appears to be highly variable due to interference caused by solids.

True Color appears to be the best indicator of pre-disinfection impact. It is possible that Soluble COD would produce similar results although that analysis was not performed. On average, the removal rate for Memphis Cellulose True Color is 27%, indicating that the majority of the True Color contributed by Memphis Cellulose passes through the treatment plant. The removal rate for Runyon True Color is 84%. The removal rate for American Yeast True Color is 72%. These three major color contributors comprise 62% of the average influent True Color received by the Stiles Facility and 65% of the pre-disinfection effluent True Color. Memphis Cellulose comprises 46% of the pre-disinfection effluent True Color due to its low removal rate which gives it an out-sized impact on disinfection. The percentage of True Color on a mass basis for the Color Contributors is as follows:

	Influent True Color	Pre-disinfection Effluent True Color
Memphis Cellulose	30%	48%
Runyon	14%	6%
American Yeast	18%	12%
All Other Users	27%	34%

The actual PAA used skews less toward Memphis Cellulose than as shown above due to excess PAA needed to buffer sudden increases in demand. Also, it has been the experience at the Stiles Facility that PAA demand is as much of a function of temperature as pre-disinfection True Color. Therefore, more PAA is consumed relative to pre-disinfection True Color in the summer compared to the winter. Other factors not currently fully understood appear to impact PAA Demand.

One indicator of the impact of Color on disinfection is the True Color: COD ratio. The below table presents these ratios related to the major color contributors for both influent and pre-disinfection effluent:

	Influent			Pre-disinfection Effluent		
	True Color	COD	Ratio	True Color	COD	Ratio
Memphis Cellulose	308	96	3.2	194	27	7.2
Runyon	148	146	1.0	24	9	2.7
American Yeast	190	0	∞	54	0	∞
All Other Users	287	778	0.4	150	75	2.0

As shown above, the True Color contributed by American Yeast received in the influent and passing through to disinfection is not associated with any increase in COD. This could indicate that the COD associated with American Yeast is very readily biodegradable but the True Color is not.

Possible Impact of Memphis Cellulose Discharge Over Time

Memphis Cellulose implemented a solids recovery project in about 2019 that greatly reduced TSS and COD in Stiles Facility effluent assuming minimum removal. See table below.

	2016-2019	2020-2023	Decrease	Percent Decrease
MC BOD ppd	37,478	19,124	18,354	49%
Effluent BOD ppd*	20,776	20,748	28	0%
MC TSS ppd	26,113	14,818	11,295	43%
Effluent TSS ppd*	20,285	9,536	10,749	53%
MC COD ppd	117,992	63,756	54,236	46%
Effluent COD ppd*	118,395	75,719	42,676	36%

*excludes 2019

It appears that the Stiles Facility picked up additional influent BOD loading after 2019 that made up for the shortfall caused by Memphis Cellulose solids recovery project.

Pre-final versus Final and Time-Weighted versus Flow-Weighted Sampling Study

Apparent Color results were compared using Time Weighted (ISCO) versus Flow Weighted sampling and at the pre-disinfection versus the post-disinfection locations. The results are shown below:

9/17/2023 to 1/12/2024				
Sample Location	ISCO	Flow composite	Statistical Comparison	p-value
Pre-final Effluent Apparent Color	631	574	Same	0.115
Final Effluent Apparent Color	653	432*	Different	0.000
Statistical Comparison	Same	Different		
p-value	0.453	0.000		

The flow composite Apparent Color result at the Post-Disinfection location was statistically lower than the other three locations. This would indicate a low bias of post-disinfection results collected via flow composite. This low bias appears to be related to configuration of the sampling pipes at just below the surface. It appears the sampling pipe intakes are located at an elevation that minimizes intake of floating and settling solids during low to average flow conditions. At higher flow levels, this does not appear to be the case. Stratification is going on at low to average flows. This had been referred to as the Golden Goose. It is very interesting that the pre-disinfection samples collected both time-weighted and flow-proportioned are

statistically the same. It is also interesting that the time-weighted samples from pre- and post-disinfection locations are statistically the same.

Biogas Investigation/Cogeneration

Fifty-three (53) samples of biogas were collected throughout the year and analyzed for siloxanes, methane, carbon dioxide, oxygen, nitrogen, hydrogen sulfide, and for BTU. Total Siloxanes as silicon averaged 78.7 ppbv ranging from 35 to 136. Hydrogen Sulfide concentrations averaged 3,609 ppmv ranging from 1,330 to 9,305. Energy values averaged 484 BTU/cubic feet (wet) ranging from 276 to 580.

Average annual values of analytical results from previous years are summarized as follows:

Year	Soloxanes (ppbv)	Hydrogen Sulfide (ppmv)	BTUs/cubic foot (wet)
2013	136	7,331	639
2014	107	4,023	602
2015	NA	3,129	570
2016	159	3,266	559
2017	131	3,083	638
2018	72	4,150	591
2019	17	3,554	588
2020	28	3,850	568
2021	57.6	9,500	464
2022	74.5	4,240	381
2023	78.7	3,609	484

The previous Stiles Facility Air Permit effective 9-16-2019 through 9-16-2024 was received on 10-2-2019. The Shelby County Health Department granted an extension of a construction permit through 8-25-2022 to allow the connection of a natural gas line to the co-gen unit. This will also involve re-commissioning the co-gen for service and then firing it with natural gas while projects are implemented to both increase biogas production and remove hydrogen sulfide. A kick-off meeting was held with Malasri Engineering on 4-14-2021 to convert the co-gen from biogas to natural gas. The Stiles Facility worked with Malasri Engineering to run natural gas to the co-gen in February 2022. A permit application for a new construction permit for natural gas fired co-gen was submitted 11-14-2022. SCHD issued the construction permit effective 3-15-2023, expiring 9-15-2024. SCHD performed annual inspection on 3-23-2023. On 8-28-2023, the new natural gas meter was finally connected to the Stiles Facility co-gen system after months of starts and stops.

The City provided comments to the draft new T5 permit on 9-22-2021. A new T5 permit was issued 3-1-2022 to expire 3-1-2027. A new permit application would need to be submitted no later than about 9-1-2026.

Mississippi River Stage Impact

Approximately 3100 acres extending from North Second Street east to Hollywood Boulevard and from James Road south to Chelsea Street is subject to backwater conditions during elevated

Mississippi River stages. River stage has major impact on Stiles Facility average daily flow beginning at 29 feet. A graph showing annual average daily flow versus number of days per year river stage exceeds 29 feet is provided in Figure 9. Total rainfall also impacts annual average daily flow. Using multivariate analysis, annual average daily flow = 0.22 x Days (> 29 feet) + 0.37 x Inches of Total Rainfall + 66 mgd from 1979 through 2023 at an R2 = 0.46. The annual frequency of occurrence is shown below:

Mississippi River Stage 1979 thru 2023			
Number of Days >29 feet	Annual Frequency	Cumulative Frequency	Interval (years)
0	31.1%	31.1%	3.2
1-10	0%	31.1%	0
10-20	13.3%	49.4%	7.5
20-30	24.4%	68.9%	4.1
30-40	6.7%	75.6%	15
40-50	6.7%	82.2%	15
50-60	4.4%	86.7%	22.5
60-70	4.4%	91.1	22.5
70+	8.9%	100%	11.3

A typical year (50th percentile) has 21 days above 29 feet river stage.

Pre-Final and Final Effluent Data Comparison—The Golden Goose

Beginning late 2017, sampling has been performed upstream (Pre-final) and downstream (Final) of the secondary treatment effluent disinfection basin. The following table presents average concentrations of samples collected:

Sample ID	Analysis	2017 Average	2018 Average (before disinfection started)	2018 Average (after disinfection started)	2019 Average	2020 Average	2021 Average	2022 Average	2023 Average
Pre-Final Effluent	BOD	43	50	44	70	42	41	27	29
Pre-Final Effluent	sBOD		28	20	36	13	18	14	15
Pre-Final Effluent	COD				256	102	150	125	93
Pre-Final Effluent	TSS	49	56	46	73	57	72	30	29
Pre-Final Effluent	Apparent Color*			815	822	460	571	539	509

Final Effluent	BOD	30	33	42	59	26	33	26	26
Final Effluent	sBOD		23	23	43	20	26	19	21
Final Effluent	COD	177	197	208	238	97	123	110	86
Final Effluent	TSS	29	31	25	40	16	15	9	11

*based on City Lab Color

The data gives an indication of the ability of the effluent disinfection tank to provide additional, post-secondary treatment removal of pollutants (i.e. The Golden Goose). Periodic disinfection began on 9-12-2018 and full-time disinfection began on 11-10-2018. After 9-12-2018, Final Effluent soluble BOD is higher than Pre-Final Effluent soluble BOD. Prior to 9-12-2018, Final Effluent soluble BOD was consistently lower than Pre-Final Effluent soluble BOD. It appears the installation of the vertical baffles to improve mixing of the injected PAA had potentially increased TSS removal after the high flow year of 2019. One mystery presented here is why Pre-final Effluent sBOD in 2018 was higher than Final Effluent sBOD prior to startup of disinfection.

Regarding the Golden Goose, in 2023, the pre-final effluent BOD average was 29 mg/l and the final effluent BOD average was 26. However, the pre-final sBOD was 15 and the final sBOD was 21 and the pre-final TSS was 29 and the final TSS was 11. Therefore, the increase in sBOD was offset by the decrease in TSS.

Laboratory

The laboratory located at the M. C. Stiles Facility received samples from the Stiles Facility and the Maxson Facility. The laboratory performed a total of 29,204 analyses, including 23,729 in-house which included 2207 E. coli tests and is compared to past years:

Year	2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
Stiles	NA	NA	NA	9355	12,953	13,019	10,089	13,157	11,775	10,322
Maxson	NA	NA	NA	7734	12,167	11.931	15,121	22,537	18,952	18,716
Chapel Hill	0	0	NA	578	603	559	549	533	447	262
Industrial Monitoring	0	0	0	0	2	0	0	339	617	514
Total	29,204	26,495	18,073	17,667	25,725	25,509	25,759	36,566	31,795	29,814

Pretreatment Program

The program regulates the discharges of all industrial users, enforces the requirements of the sewer use ordinance, implements the federal categorical standards on specific process wastewaters, and manages the hauled waste program. In 2023, as part of regulating these industrial users, the Control Authority collected 605 samples, performed 91 site inspections, and issued 206 Notices of Violation. There were 91 significant industrial users (SIUs) that were monitored and reported on to the Approving Authority (TDEC). A summary of such statistics from previous years is as follows:

Year	2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
Samples	605	134	558	153	295	285	244	313	205	181
Inspections	91	92	93	92	105	107	98	105	93	101
NOVs	206	179	220	223	195	233	220	288	146	206
SIUs	91	92	93	95	99	104	101	98	98	95

For the year 2023, the City collected $208,246 for disposal of 1,367,645 gallons of hauled wastewater. Most of the hauled waste deliveries are now septic related although this was not the case prior to 2023. A summary of such statistics for the last 10 years is as follows:

Year	Rate/gallon	Volume (gallons)	Revenue
2023	13 cents	1,367,645	$208.246
2022	13 cents	550,300	$71,539
2021	13 cents	1,203,955	$199,566
2020	13 cents	2,230,824	$357,101
2019	13 cents	2,416,765	$365,222
2018	13 cents	4,818,131	$626,357
2017	13 cents	4,649,995	$604,553
2016	13 cents	5,009,616	$651,160
2015	13 cents	8,550,823	$1,111,607
2014	13 cents	9,319,286	$1,211,507

The sewer fee history is summarized below:

Date	Volumetric	BOD Surcharge (250 mg/l)	TSS Surcharge (300 mg/l)	COD Surcharge (800 mg/l)	Ordinance
5-15-1979	28.52 cents	1.7 cents	0.9 cents	-	#2877
7-1-1981	-	3 cents	5 cents	-	#3127
1-1-1982	58.68 cents	2.7 cents	4.6 cents	-	-
7-1-2004	87.5 cents	3.65 cents	5.96 cents	-	#5036
7-1-2008	98.5 cents	3.86 cents	6.21 cents	-	#5239
7-1-2009	105.3 cents	-	-	-	#5312
8-1-2009	-	3.88 cents	6.61 cents	-	-
7-1-2010	226.7 cents	4.28 cents	7.16 cents	-	#5356
1-1-2018	287 cents	-	-	-	#5651
1-1-2020	332 cents	-	-	3.96 cents	#5651
1-1-2021	-	-	-	4.821 cents	#5651
1-1-2022	-	-	-	4.821 cents	#5651

The Maxson Facility rated capacity is 90 mgd for flow, 490,000 ppd for influent BOD and 410,000 ppd for influent TSS. The Stiles Facility rated capacity is 135 mgd for flow, 275,000 ppd for influent BOD and 374,000 ppd for influent TSS. Influent flow and loading trends are shown in Figures 10-12 for the Maxson Facility and in Figures 13-15 for the Stiles Facility. For the Maxson Facility, Figure 10 shows annual average daily flow and average daily dry weather flow since 1976. Figure 11 shows annual average influent and industrial BOD loading

since 1976 and 1989 respectively. Figure 12 shows annual average influent and industrial TSS loading since 1976 and 1989 respectively. For the Stiles Facility, Figure 13 shows annual average daily flow and average daily dry weather flow since 1978. Figure 14 shows annual average influent and industrial BOD loading since 1978 and 1992 respectively. Figure 15 annual average influent and industrial TSS loading since 1978 and 1992 respectively.

Since 1979, the Additional Treatment Cost charged to industries has increased at an annual rate of 2.09% for BOD and 4.76% for TSS and the volumetric charge for all users has increased at an annual rate of 5.65%.

The COD surcharge was intended to cover 50% of disinfection cost. Below is a summary of annual estimated COD surcharge revenue and disinfection cost:

Year	Estimated COD Surcharge Revenue	Disinfection Cost	Percentage covered by COD Surcharge
2019	$11,067,792	$14,075,692	85%
2020	$11,088,780	$11,519,858	107%
2021	$16,092,865	$18,389,255	98%
2022	$14,035,460	$22,267,852	64%
2023	$11,522,083	$21,739,780	53%

3.0 Analysis and Future Outlook

Analysis

On 1-10-1967, the City Commission adopted a resolution urging the State of Tennessee to participate in a sewage treatment program under the upcoming Federal Clean Water Act of 1968 for the construction of sewage treatment facilities. Sewer and industrial waste sampling commenced in 1967 as part of 201 Facilities planning under MATCOG and the US EPA. A flat monthly sewer charge for residential and other users was adopted by the City in July 1969 to provide local funding for these anticipated projects. On 2-5-1969, the Memphis Water Pollution Advisory Committee issued a statement that Memphis industries may have to shut down unless the City moves to clean-up its "open ditch" sewer system. On 4-8-1969, the City Council considered grants for the construction of two major wastewater treatment facilities as well as a new interceptor sewer for the Nonconnah Creek Basin. On 1-6-1970, the City Council adopted a resolution setting a construction schedule for secondary treatment facilities with the south plant to be completed by July 1974 and the north plant completed by December 1978. Contracts were negotiated with two joint-venture engineering teams to design and oversee construction of each treatment plant. Allen & Hoshall and Clark, Dietz & Associates were selected to design and oversee the construction of the south plant. Ellers, Reaves, Fanning, and Oakley and Consoer Townsend were selected for the north plant. Both design teams embraced contact-stabilization to meet the requirement for secondary treatment without the use of primary clarifiers as well as chlorine disinfection. The design, start up, and operation of each plant, however, followed different tracks and had different challenges.

The Maxson Facility treatment capacity history is summarized as follows:

Year	Flow (mgd)	BOD (ppd)	SS (ppd)
1975	80	186,000	133,500
1985	80	390,000	335,000
1997	90	490,000	410,000

The City purchased the land on 10-8-1971 for $65,000 from Shelby County to construct the Maxson Facility. In 1972, with the passage of P.L. 92-500, nationwide 30-30 effluent discharge limits were established for publicly owned treatment works constructed and in operation before 7-1-1977. The design criteria pre-dated the 1972 legislation that only required 85% removal.

The Maxson Facility was issued NPDES permit effective 11-30-1974 and expiring 12-31-1979. For the first two years of the permit, percent removal limits of 85% for BOD and TSS would be applied. Starting 1-1-1977, the 30-30 effluent concentration limitations would be applied. The plant officially opened on 6-4-1975 to much fanfare after having been in operation for about a week. Records from November 1975 through June 1977 indicate influent monthly average loadings approaching 300,000 ppd for BOD and 250,000 ppd for TSS and monthly average removals of 80% for BOD and 62% for TSS.

On 7-22-1977, the U.S. DOJ/EPA filed suit against the City for violations of the discharge permit from 6-30-1975 through 7-22-1977. The Court entered the Consent Decree on 7-28-1978

to increase plant capacity. The Decree was modified on three subsequent occasions: Amended Decree (2-23-1979); Second Amendment to Decree (5-7-1979), and Consent Order (12-31-1981). The work was performed utilizing ten contracts consisting of additional blower purchase (Blower #6), piping modifications, blower installation, electrical modifications, site preparation, three (3) primary clarifiers, administration building modifications, electrical reliability modifications, computer modifications, and secondary treatment modifications (4 ABF Towers, 5 Bio-pumps). The Decree was terminated on 10-10-1986.

The City elected to make the required improvements using EPA Construction Grants. The City received approval for its grant application from the EPA on 12-20-1979. The City entered into a Patent License and Consulting Services Agreement with Neptune Microfloc, Inc. in 1981 for the rights to use the activated bio-filter (ABF) process. The City obtained an Innovative Technology Grant on 3-8-1982 to utilize the ABF process as the first stage secondary treatment of a high strength (600 mg/l BOD5) wastewater. The EPA approved the funding of 4 instead of 6 bio-filters as originally requested taking the stance there was sufficient capacity through 1985 loading projections provided the bio-filters were fully loaded.

The Maxson Facility went on-line in March of 1986 with all 4 bio-filters in service. An Evaluation of Operating Data was performed on data collected from May 1985 to February 1986. The report showed that the plant would have trouble staying under 30 mg/l for effluent BOD and TSS during the colder months. On 10-24-1986, the City proposed a Supplement to Evaluation of Operating Data from the T.E. Maxson Wastewater Treatment Plant to the TDHE Division of Construction Grants. One bio-tower would be taken off-line to allow the other three to be fully loaded and then operated during the coldest time of the year. The evaluation was performed from December 1986 through February 1987. During this period, three bio-filters were operated with 4 bio-pumps, three bio-filters were operated with 3 bio-pumps, and two bio-filters were operated with 3 bio-pumps.

A review of operating data from December 1986 to February 1987 confirmed that the organic removal across the towers is a function of the transfer capability within the bio-towers which depends on the temperature of the wastewater and the hydraulic loading rate. The evaluation concluded that at maximum month design loads, the biotreatment system would fail completely and EPA effluent limitations would be exceeded. Among other alternatives, the report recommended adding two additional bio-pumps to increase the overflow rate and converting the 4 existing aerobic digestion tanks to regular aeration.

A letter from TDHE Construction Grants and Loans dated 2-29-1988 to the USEPA indicated the City had sent a modification request respective to an innovative design application and concluded that the upgrade would not consistently operate as designed. TDHE was transitioning to the State Revolving Fund program and existing M/R funds were dissipated so they requested EPA review the request in a timely manner. The M/R Grant was approved, and the construction performed 1989-1990.

The Maxson Facility existing primary clarifier sweeps were modified during 1991-93 to increase sludge removal. A second capacity expansion took place 1995-1997 consisting of the addition of two ABF Towers, a 4th Primary Clarifier/Thickener, and two Secondary Clarifiers. A covered lagoon capacity expansion project was completed in 2004 which covered Lagoons 2A, 2B, and 2C and improved solids capture and primary sludge pumping. Primary sludge dewatering was discontinued in 2004 and the wasting of primary sludge directly to the lagoons for co-digestion with waste activated sludge began. In 2006, the bio-towers began periodic operation in roughing

mode resulting in energy savings, improved effluent quality, and a dramatic reduction in waste activated sludge. This practice continued sporadically through the second half of 2019 when discontinued due to valve problems. In 2010, a new cover was put on Lagoon No. 1 and the influent head works replaced. In 2013, new Fine Bar Screens were put into operation, reducing plastic items wasted to the lagoons. The Primary Scum Removal System was reworked in 2014 increasing the overall removal efficiency of the primary clarifiers.

Co-digestion of primary sludge with waste activated sludge beginning in 2004 along with covered lagoon expansion led to a dramatic increase in biogas production that continued through 2016. In 2017, biogas production began falling off. There had been a decrease in activated sludge wasting in recent years due to increased use of roughing mode contributing to the decline in biogas production. In addition, inert solids build up in Lagoon 2A may have accelerated after the TVA ash release in 2010 contributing to the decline in biogas production.

The expected unmonitored influent BOD should be 69,000 ppd based on population size (an estimated 404,570 people) or 92,000 ppd based on minimum flow and 250 mg/l BOD. Figure 16 shows trends in unmonitored influent BOD. The increase in BOD at the head of the plant was totally unexpected after completion of major lagoon improvement projects which focused on solids retention.

Similarly, the expected unmonitored influent TSS should be either 81,000 ppd or 110,000 ppd based on minimum flow and 300 mg/l TSS. Figure 17 shows trends in unmonitored TSS. The monitored result includes the 24-inch dewatering filtrate / lagoon supernatant sampling point. Two other return flow lines including Lagoon No. 1 and Horn Lake Ditch are not monitored. The graph shows major spikes in unmonitored influent TSS occurring since 2004. The cause of these spikes can be attributed to unmonitored lagoon returns and unrepresentative influent sample collection.

Average annual effluent BOD concentration since 1975 is shown in Figure 18. The figure shows gradual increase related to increases in influent loading for a 20-year period beginning in 1987. For the last 10 years or so, increases appear to be due to worsening removal efficiency.

Average annual effluent TSS concentration since 1975 is shown in Figure 19. The figure shows gradual increase related to increases in influent loading beginning in 1987. The increases over the last 10 years or so appear to be due to worsening removal efficiency.

As shown in Figure 20, combined primary and secondary sludge wasting peaked in 2006. Beginning that year, as previously discussed, the bio-towers began to be periodically operated in roughing mode for the first time producing less waste activated sludge. The Maxson Facility has been operated mainly in ABF mode since after mid-2019. As shown in Figure 21, annual power consumption peaked between 1997 and 2004. Power consumption began decreasing in 2005 as the plant began wasting primary sludge directly to the lagoons and operating more in roughing mode beginning 2007 resulting in greater efficiency of the bio-towers and less blower operation as well as improved effluent quality. This downward trend continued until 2010. Energy consumption has been increasing since then due to inability to maintain consistent roughing mode among other conditions that tend to increase blower usage despite consistent influent loading.

The City contracted with the joint venture for engineering services on 10-31-1969 to begin design of the Stiles Facility originally known as the North Sewage Treatment Plant. The City

acquired the site location in 1971 consisting of 208 acres, about 86 acres of which are located on land fill produced by the spoil of the ACOE dredging work in stabilizing the Mississippi River and Wolf River channels in 1965. The surface elevation of the land fill area where the main plant facility is situated varies 240 to 242 feet AMSL. Prior to dredging, the natural surface area elevation varied 210 to 225 feet AMSL. Approximately 20 acres of additional land fill surface area adjacent the Mississippi River was deeded to the Lucy-Woodstock Utility District for barge docking and chemical supply station for an industrial pipeline. The area immediately north of the main plant facility was acquired by the Airport Authority for relocation of a small municipal airport formerly located on the southern end of Mud Island. The airport was constructed and opened to the public on 5-30-1971. A report "Basic Design Data for Preparation of Contract Drawings" was issued June 1971, contained the basic parameters for plant operations. The Stiles Facility was issued NPDES permit effective 11-30-1974 and expiring 12-31-1979. For the first two years of the permit, percent removal limits of 85% for BOD and TSS would be applied. Starting 1-1-1977, the 30-30 effluent concentration limitations would also be applied.

The Stiles Facility began operation in August 1977. By January 1979, the plant was still not consistently meeting its 30-30 effluent concentration limitations. CH2MHill was hired to evaluate operating data and to determine the maximum treatment capabilities of the plant it its present configuration at existing and future loading rates. The report issued April 1980 concluded that major plant modifications would be required (i.e. adding primary clarifiers) if future influent loading conditions (i.e. 450,000 ppd BOD) increased above its current design load (275,000 ppd BOD) assuming that aerobic digestion was eliminated. Full scale operational testing showed that the plant could meet 30-30 standards at design loading by operating 8 contact and 10 stabilization tanks. It was further recommended that the permit limits be raised to 60-60 due to the high industrial contribution to the plant.

As discussed, the design, start-up and operation of the Stiles Facility has been relatively straight forward with no major process modifications since original construction. Its original design reportedly was based on the Stinkney Plant in Chicago. The aerobic digestors were discontinued in the early 1990s. Industries have come online in recent years to bring the plant up to full treatment capacity. In 2003, American Yeast began production in February and KTG began production in March. Discharges increased as these facilities ramped up to full production. In 2013, KTG expanded production with the startup of its TAD unit. Runyon's evident expansion of operations in 2021 in an apparent sneak attack pretty much topped off the Stiles Facility BOD tank along with other refractory qualities.

Trends in unmonitored influent BOD loading are shown in Figure 22. The expected unmonitored influent BOD should be 73,000 ppd based on population size (an estimated 427,435 people) or 85,000 ppd based on minimum actual flow and 250 mg/l BOD. Until the last couple of years, the lowering and increased stability of that number would indicate that industrial discharges are being effectively monitored for that parameter. However, unmonitored BOD has been increasing to a level of concern. Unmonitored influent TSS has historically been erratic and higher than expected over the years but has been becoming steadier and trending upward as shown in Figure 23. The expected unmonitored influent TSS based on population this size (an estimated 427,435 people) is 85,000 ppd or 103,000 ppd based on minimum actual flow and 300 mg/l TSS.

Average annual effluent BOD concentration since 1978 is shown in Figure 24. Since the plant began operation in 1978, effluent BOD concentration has been a function of influent loading until after 2019 when there was a sudden decrease in concentration apparently related to Memphis Cellulose installing a fiber recovery unit.

Average annual effluent TSS concentration since 1978 is shown in Figure 25. Over the last 20+ years, effluent TSS concentration has been a function of influent loading with the relationship being one of general trend direction and not one-to-one.

As shown in Figure 26, sludge wasting peaked in 1988 and was subsequently on the decline for many years reaching a low point in 2003 when both American Yeast and KTG both began production. Sludge production then picked up mirroring the discharges of these two industries. Presumably, American Yeast's high strength organics was converted to solids and wasted from the activated sludge process and KTG's mainly very low volatile solids were direct wasted to the lagoons. After the economic slowdown, sludge wasting began picking back up in 2014, helped by the increase in influent solids loading after the addition of the TAD unit at KTG. As shown in Figure 27, energy consumption was on a 20-year decline starting in 1981 as the plant implemented the Aeropt system and discontinued aerobic digesters. Energy consumption began increasing in 2003 when both American Yeast and KTG began production. Aeropt was discontinued in 2010 and energy consumption remained flat for several years. Energy consumption has increased in recent years due to increased blower usage to improve effluent quality, especially upstream of PAA addition and deterioration evident in secondary clarifier weirs. The decrease in energy consumption in 2023 was caused by blower problems brought on by summer thunderstorms.

The cost that MLGW charges for electricity was in the news for a while with talk about MLGW severing ties with TVA. Essentially, the wastewater treatment plants pay a wholesale rate from TVA. The average annual all-in cost for main power going back to 1996 is shown in Figure 28. The 2023 wholesale rate for power by the City's wastewater treatment plants was 8.02 cents per kwh. The MISO wholesale rate for power generation in June 2023 was around 3.7 cents which was a 53% reduction from 2022. This indicates that the City may not always be paying the lowest available rate.

Table 1 and Table 2 both show influent concentration and loading and effluent concentration for BOD, COD, and TSS over the years. Table 3 compares total influent flow, average daily dry weather flow, and average daily billed wastewater flow at both treatment plants compared with drinking water production for each month. The total influent flow is the average daily flow received at the treatment plants. The average daily dry weather flow is based on days with no apparent river or rain impact. The average daily billed wastewater flow is based on residential and commercial water usage and billed large industrial user and wholesale customer sewer usage. Drinking water production is the metered output of the water treatment plants that is sold to residential and commercial customers and a small portion of it is not affiliated with the City. Several industrial users use private wells to generate cooling and process water instead of using MLGW drinking water which is discharged to the sanitary sewer on a continuous basis and may constitute 10% or more of dry weather flow received. Figure 30 shows the Table 3 data over the last 10 years.

Tables 4 and 5 present average annual rain-induced flow per inch of rain, percent of average daily flow that is rain-induced, average daily flow, average daily dry weather flow, annual rainfall, and percent of average daily flow that is river-induced for each treatment facility since 1975 and 1979 for the Maxson and Stiles Facilities, respectively. Trends in rates of rain-induced flow are shown in Figures 29 and 30. Over the last 10 years, rain-induced flow has averaged 28 million gallons per inch of rain at the Maxson Facility and 27 million gallons per inch of rain at the Stiles Facility. Trends in average daily and average daily dry weather flow for both facilities combined are shown in Figure 31. Overall, rain-induced flow has comprised 6% and 7% of average daily flow received at Maxson and Stiles, respectively, and river-induced flow has comprised zero % and 5% of average daily flow received, respectively. Combined ADF peaked in 1998, reached its lowest point in 2006, and climbed to a new peak in 2019. Combined ADDWF peaked in 1996 and was on steady decline through about 2016 and has been slightly increasing since then whereas billed daily flow as shown in Figure 32 has been steady for the last 10 years.

Data related to plant operations and sludge production is presented in Tables 6 and 7. Some additional tables presented are Table 8, the monthly plant operating budget expenditures and Table 9, a breakdown of plant expenditures by category. Treatment costs averaged 119 cents per 1000 gallons in 2023, an increase of 8 cents from the previous year as shown in Figure 33 with previous years for comparison. Annual cost to treat infiltration / inflow are shown in Figure 34. These costs are more significant now that disinfection of secondary effluent has started. Tables 10 and 11 comprise the largest industrial user's monthly fees for each plant. Table 12 shows the monthly fees of the three wholesale customers. These tables show total fees of $49,113,600 from these 19 industrial users, hauled wastes, and 3 wholesale customers. If you deduct the portion of disinfection cost not covered by the COD Surcharge or $10,216,757 from the treatment plant O&M costs, then the fees from the 19 industrial users, hauled wastes, and 3 wholesale customer fees covered 91% of the costs. The total fees received decreased $3,692,402 for the 19 industrial users and decreased $296,728 for the wholesale customers for a total decrease of $3,989,130 for these revenue categories from the previous year. Table 13 shows annual hauled waste volume and revenue over the years.

Table 14 is a monthly comparison of the utility charges for both treatment plants. The treatment plants changed to Time of Use contracts effective 10-1-2015 which makes unit cost tracking more complicated due to different on-peak and off-peak time blocks.

Table 15 is a mass balance of the origin of the total suspended solids in the Maxson Facility influent. This table shows that the lagoon supernatant was 12% of nominal plant TSS loading in 2023. Table 16 shows the total loading of BOD_5 of the plant and the amount contributed by the major industries. Table 17 shows the total loading of COD of the plant and the amount contributed by major industries. Table 18 shows the total loading of total suspended solids of the plant and the amount contributed by the major industries. Table 19 shows the water usage and cost. Table 20 is a table showing the electrical usage, cost, biogas production, and propane usage for the covered lagoon. Table 21 is a table showing the electrical usage and cost for other electrical meters. The City uses an average of 1.4 mgd of potable water at the Maxson Facility for cooling equipment and dewatering spray down and could be asked to transition to a lower quality water source based on future industrial need and potential new regulatory requirements. Table 22 shows long-term trends in BOD loading and wasting.

Table 23 shows the utility usage, cost, and biogas production for the Stiles Facility. Table 24 shows the plant water usage and cost as well as the electrical usage and cost for dewatering. Table 25 shows total loading of BOD_5 for the Stiles Facility and the amount contributed to it by the major industries. Table 26 shows the total loading of COD of the plant and the amount contributed by major industries. Table 27 shows the total loading of TSS for the plant and the amount contributed by the major industries. The City uses an average of 0.3 mgd of potable water at the Stiles Facility for cooling equipment and dewatering spray down and could be asked to transition to a lower quality water source based on future industrial need and potential new regulatory requirements. This average appears low and should be closer to 1 mgd and indicates possible problems with the meter. Biogas purchases by American Yeast are shown in Table 28.

Table 29 summarizes the electrical usage, total cost, unit charges, and savings over the past several years for both treatment facilities. Table 30 and Table 31 show comparisons of the Stiles Facility and Maxson Facility influent flows versus the combined flows of the major industries.

Recent Regulatory Developments

Final NPDES permits for both treatment plants were issued on 12-2-2011 for the Stiles Facility and on 12-6-2011 for the Maxson Facility, effective 1-1-2012 and expiring on 12-31-2016. Certain elements of the permits were appealed. TDEC proposed a modification to the Maxson Facility permit on 9-10-2012 based on comments received on the draft permit public noticed on 4-9-2009 and re-public noticed on 10-11-2010 as part of settlement negotiations related to the permit appeal. The proposed modified permit rationale recognized the variance from federal treatment standards, the deletion of mass limits for effluent parameters (like Stiles), and the deletion of effluent flow metering pending a flow study. However, TDEC did not pursue the matter further. Applications for new permits were submitted 7-5-2016. The City met with TDEC on 4-10-2017 to review the issuance of new permits. TDEC presented information at the meeting indicating the requested secondary treatment adjustments (i.e. variances) would not be granted. On 10-20-2017, the City submitted a report explaining why the variances should be granted as a supplement to the previously submitted permit applications. New draft permits for both treatment plants were public noticed on 11-13-2018. The permits stuck to the nationwide 30-30 standards and did not allow Best Professional Judgement (BPJ) to be used for industry groups without promulgated categorical effluent limits for secondary treatment adjustment. Public Works officials drove to Nashville on 12-10-2018 to meet with TDEC and learned that TDEC was not flexible on this issue. The City was granted an extension request for public comment to 2-15-2019. A public meeting for the Maxson and Stiles Facility proposed NPDES permits was held on 2-26-2019. TDEC would not impose a permit limit on PAA residual since the City would have an incentive to keep it low due to cost. The City submitted comments on 3-20-2019, totaling 800+ pages. On 4-18-2019, Vojin Janjic emailed that based on the comments received both NPDES permits would be withdrawn, and a new draft permit issued later which would require a new public notice. As the permit negotiations continued, in early 2021, TDEC legal personnel concluded that the state did not have the authority to implement secondary treatment adjustments without additional language in the regulations. On 4-20-2021, the Tennessee Water Quality Board approved the City's proposed amendments to the state regulations that would allow adjustments to TBELs in accordance with 40 CFR 133.103 (b). The amended regulations were promulgated on 5-15-2022. On 10-14-2022, the City submitted the most recent industrial user data in response to TDEC request related to Secondary Treatment Adjustments for the next NPDES permits. Previous NPDES permit limits for treatment and performance are shown in Table 32. As shown in the table, secondary treatment adjustments of effluent limits were required for compliance and have varied, getting somewhat looser over the years.

The new NPDES permit was public noticed on 7-19-2023. On 8-17-2023, the City made a request to TDEC for a 90-day extension from 9-9-2023 to 12-7-2023 to provide NPDES permit comments.

The public hearing for the NPDES permits was held at the Main Library on 8-28-2023 with written comments due 12-7-2023 and final permits expected to be issued after Christmas. At the meeting, TDEC emphasized the permit limits being technology and not water quality based. There was also interest expressed by some in the audience regarding effluent re-use which TDEC indicated was covered by the regulations. The proposed versus current permits limits for BOD and TSS are shown below:

Maxson Facility	Monthly Average	Weekly Average	Daily Max
Proposed BOD	34.8	52.2	51.1
Current BOD	42	63	84
Proposed TSS	37.8	56.8	54
Current TSS	48	72	96

Stiles Facility	Monthly Average	Weekly Average	Daily Max
Proposed BOD	53.8	80.7	107.6
Current BOD	43.1	64.7	86.1
Proposed TSS	40.7	61.1	81.4
Current TSS	53.2	78	104

City comments were submitted on 12-7-2023. The main concern were the "rules" developed by the permit writer in the rationale for application of the federal secondary treatment adjustment regulations in the most stringent manner and the definition of reasonable measure of industrial production. The amended Uniform Administrative Procedures Act which was enacted in 2022 was cited stating that a court appeal would not consider a state agency's interpretation of the statute or rule.

Demonstrating compliance with 40 CFR 133.102 as it relates to treatment and performance has been a concern since the implementation of PAA disinfection beginning late 2018. The Stiles Facility compliance point for BOD and TSS treatment and performance has been via a flow weighted composite sample of final effluent collected via two horizontal pipes about 4 feet below the surface on each side of the disinfection basin. This sampling apparatus dates to the original construction of the treatment plant. To provide continuity of data, it was decided to request a BOD credit calculated on PAA dose and keep everything else the same. The concept of BOD credit appeared to be reasonable based on three studies performed on Stiles Facility effluent by vendors and consultants over the years. However, all were based on grab samples. The Stiles Facility initially used portable Isco autosamplers and later installed a permanent pre-final effluent sampling apparatus (going into service on 9-17-2021) comparable to the final effluent sampling apparatus and embarked upon a formal study, including measuring the increase in soluble BOD across the disinfection tank due to PAA addition. The study revealed additional BOD and TSS removal was occurring across the disinfection tank during the monitoring period (2017-2021) mainly due to additional TSS removal that also lowered BOD. The question now is why select solids are not being captured in the final clarifiers. Getting this issue resolved is

very important, especially if the pre-disinfection sampling point is ever used for secondary treatment compliance.

For the Maxson Facility, more than likely, secondary treatment will be based on the existing final effluent sample collected upstream of the disinfection tank and PAA impact on BOD would not be considered.

The third Maxson Facility expansion project was initiated in 2014 at the encouragement of City officials to facilitate expansion of existing industries as well as establishment of new industrial development in the Pidgeon Industrial Park. During the early years of the City of Memphis Pretreatment Program, discharge agreements were signed with industrial users and industries sought to "negotiate" greater discharge limits than currently needed for future expansion. In later years, the City moved to a permit approach which had less perceived opportunities for "negotiation" and began to reign in past promises to industries in an informal effort to avoid over allocation compared to rated treatment capacity. As such, industries began contacting the Mayor's office to secure such commitments outside of the industrial user permit system. For the moment, an informal truce has been reached between existing industries and the wastewater treatment plants with both treatment plants near capacity and able to meet current permit limits. An expansion would allow existing industries to be able to plan for future expansion and allow the City to recruit new industries to the Pidgeon Industrial Park where there is plenty of land and other resources available.

The expansion project initiated in 2014 was broken into five smaller separate GMPs which, taken together, would encompass the desired capacity increase and performance improvements as well as disinfection and influent coarse bar screen replacement. The cost to perform this work was slated to be approximately $251 million with approximately $83 million funding from SRF.

The Maxson Facility upgrades will go a long way toward ensuring compliance in a regulatory scenario that is worst case (i.e. 30-30 limits) especially if rated capacity becomes too large for any one industrial category to qualify for the 10% rated capacity threshold. The Stiles Facility is currently loaded to capacity and should benefit from the secondary treatment adjustment based on its current industrial mix.

On 11-27-2007, the Natural Resources Defense Council petitioned the US EPA to request updated information about secondary treatment nutrient removal capability and to establish new technology-based nutrient limits as part of the secondary treatment standards. On 3-16-2011, a Nutrient Reduction Framework (NRF) memo was issued to the EPA regional offices to guide EPA technical assistance and collaboration with states. This initiative was subsequently rolled down to the states. On 12-14-2012, the EPA responded to the NRDC petition saying that it agreed that insufficient data exist to draw any general conclusions about the ability of secondary treatment to remove nutrients and that a uniform set of nationally applicable, technology-based nutrient limits was not warranted. In February 2016, the EPA issued the 1st Gulf of Mexico Hypoxia Task Force report. On 10-16-2019, EPA issued the 2nd Gulf of Mexico Hypoxia Task Force report. Both the Maxson and Stiles Facilities were listed as major point source contributors from Tennessee. On 8-17-2020, it was announced that HTF would begin working with SERA-46 and other research groups to evaluate factors that contribute to nutrient loss from agricultural lands, state-level nutrient impairments, and hypoxia in the Gulf of Mexico. In 2021,

the HTF announced potential new monitoring sites in the Mississippi-Atchafalaya River basin, particularly to track loads and trends in large rivers to help states evaluate progress toward meeting nutrient reduction goals and to support adaptive management of nutrient reduction strategies. The 4th Gulf of Mexico Hypoxia Task Force report was issued in October 2023. The HTF remains committed to its 2035 goal of reducing the 5-year average areal extent of the hypoxic zone in the Gulf to less than 5,000 square kilometers by 2035, with an interim target for reducing total nitrogen (TN) and total phosphorus (TP) loads by 20 percent by the year 2025.

In addition, the EPA initiated a multi-year, multi-phased national study of nutrient discharges from POTWs to 1) establish a statistically representative, nationwide baseline for nutrient discharge and removal at these facilities, and 2) characterize operational and management practices that result in improved nutrient reduction. A questionnaire was published in the Federal Register on 9-19-2016 but was withdrawn after the presidential election. The City responded to a POTW Nutrient Survey / Voluntary Questionnaire issued by the EPA on 10-22-2019 for both treatment facilities. The EPA compiled the survey responses collected from 2019 through 2021. Survey results to date show more than 1,000 POTWs with different biological treatment types (including both conventional and advanced treatment technologies) can achieve effluent total nitrogen of 8 mg/L and total phosphorus of 1 mg/L. In 2023, the EPA continued to support state programs and making funds available to POTWs through SRF and IRA.

In 2014, TDEC submitted a draft NRF to the EPA for review and subsequently began implementing it. The TDEC NRF prioritizes HUC10 watersheds with 303(d) listed segments. Tennessee is developing a Non-Point Source nutrient reduction strategy for Mississippi River basin (Hydrogeological Unit IV) using a farmer-led approach so it is not known at this time how the Maxson and Stiles Facilities will be affected. The initial state-wide implementation of nutrient reduction would emphasis optimization instead of capital improvement. In 2018, TDEC's emphasis was on developing a larger stake holder group for input on the NRF in general and developing site-specific criteria for three reservoirs in Tennessee as a priority.

At the direction of the governor, in 2019, TDEC and TDA convened the Tennessee Nutrient Strategy Taskforce (the Taskforce), drawing representation from academia, state and local government, wastewater treatment plant operators, the private sector, and non-governmental organizations. In 2021, the Tennessee General Assembly issued joint House Resolution 751 recognizing the work of Tennessee representatives on the Hypoxia Taskforce. Among other objectives, the Taskforce planned to generate a list of municipal wastewater treatment plants in Tennessee with permits for: TN and/or TP limits, Report TN and/or TP or No Reporting required and, if possible, planned to develop a consolidated spreadsheet with flow, loads and nutrient concentration data for wastewater treatment plants. During 2020-2022, a tool kit was developed by Middle Tennessee State University by EPA grant for use by stake holders. The tool kit is not yet final.

WEF announced on 2-2-2022 a "Nutrient Smart Recognition Program" in collaboration with the EPA. Wastewater treatment facilities would submit on a voluntary basis loading and discharge data for nitrogen and phosphorus as part of a 1-year pilot program. Facilities achieving a 90% removal would receive a Platinum Award as well as lessor awards and awards for innovation.

In 2022, TDEC announced it was starting a new round for its Tennessee Plant Optimization Program (TNPOP) for nutrient reduction. Instead of a state-wide program for nutrients, TDEC

indicated it would take matters on a case-by-case basis, including establishing criteria for reservoirs. Federal money would available for states participating on the Gulf Hypoxia Task Force as well. In 2023, the Taskforce continued to meet. A goal was set for the TNPOP to have reached all POTWs by 2030. At the WPC23, TDEC stated that POTWs should use Local Limits to control nutrients, including PFAS.

Corrosion-The Silent Killer

Coupons were installed at the Maxson Facility from April through May 2020 to assess current corrosion rates and project future corrosion conditions. The areas surveyed were rated "Severe" except for the main control room which was rated "Harsh." The Severe designation means that electronic/electrical equipment is not expected to survive. Each area represents a unique challenge especially as the City transitions to SCADA which involves more delicate hardware with sensitive metals designed for modern control room or data center atmospheres not typically encountered at wastewater treatment plant environments. Testing at the Stiles Facility has produced similar results. Corrosion is an equal concern in both POTW and Collection System infrastructure.

Metals

The Pretreatment Program had set allowable limits for many categorical dischargers that resulted in the significant reduction of metals received over the past three decades. With the promulgation of the 503 Sludge Regulations, the various metals entering the plants became a greater regulatory concern. Prior to 1994, the discharge of metals had been controlled and regulated but there were no regulations that defined allowable concentrations of various metals in biosolids. Table 33 shows the metals concentrations over the years in the plants' influent and biosolids. The tables utilize the combination of the six most common metals, zinc, copper, nickel, chromium, lead, and cadmium. As the table shows, influent metals loadings have generally decreased over the past 4 decades.

Table 33 shows the average limit that would apply to the Maxson and Stiles Facilities either as land application or surface disposal under the 503 regulations as well as the metals concentrations for biosolids sampled in 2022. The metals are arsenic, cadmium, chromium, copper, lead, mercury, molybdenum, nickel, selenium, and zinc. Metal concentrations in surface-disposed bio-solids have decreased over the past 3 decades at both treatment plants, although at a slower pace compared to influent metals.

With the introduction of land application of biosolids in 2018 for the Stiles Facility, the frequency of sampling of biosolids for metals has increased above the bare minimum requirements listed in the NPDES permits indicating seasonal variation and possibly episodic increased contribution of metals by industrial users previously unnoticed. Tables 34 and 35 representing monthly metals results for the Maxson and Stiles Facilities, respectively, are introduced in this annual report.

More frequent influent sampling as well as specific interceptor sampling (i.e. Nonconnah Interceptor) is warranted as a proactive measure to evaluate any trends and potential specific sources based on sampling, especially related to Molybdenum which has a lower threshold, 75 mg/kg, for regulatory concern.

Personnel

Previously, all but one of the classifications represented by 369D feature apprenticeship-like step raises based on time in position and financial incentives for certifications. Except for craft unions, many non-exempt employees at the treatment plants came in at the lowest classification and promoted based on most qualified applicant.

In 2011, 369D negotiated a clause in the MOU stating the permanent filling of any vacancy be promotional only to the senior most qualified applicant, whenever possible, except those classified as entry level positions as designated by the Director of Public Works and replaced incentive pay with automatic promotion for maintenance mechanics from M1 to M2. As explained later, this seemingly curtailed management's ability to hire external candidates for the M2 position. Public Works hired a consultant in 2011 to perform a job analysis showing that the Mechanical Operator was not entry level for either the wastewater treatment operator or the maintenance mechanic career paths.

In 2013, Public Works created its own apprenticeship-like program conforming with the state wastewater treatment operator certification requirements. The program consisted of non-union trainee position with a 36-month probationary status that would act as an entry point to the regular wastewater treatment operator position. The obtainment of a Grade 4 certification was the condition for continued City employment as a regular wastewater treatment operator.

The City has elected in the past not to participate in joint apprenticeship and training committees as defined by the U.S. Department of Labor or contributing to any union training funds. The recruitment of maintenance mechanics is hampered by lack of applicants that meet minimum qualifications for journey level or graduate from vocational schools (i.e. pre-apprenticeship program) and the inability to recruit more experienced mechanics with specialty credentials at higher pay due to union restrictions (i.e. Maintenance Mech 2 is promotion-only due to union agreement).

The Workforce Innovation and Opportunity Act (WIOA), signed into law on July 22, 2014, promotes the use of Registered Apprenticeship programs. The State of Tennessee created Apprenticeship Tennessee in October 2020 to work with employers and local educational institutions to develop and expand the use of federally registered apprenticeship programs in the state. Public Works has the option to participate in a registered apprenticeship program, develop a registered program of its own, contribute to a union training fund, or create its own apprentice-like program conforming to its own unique requirements.

In September 2021, HR approved a revised salary progression plan to allow wastewater treatment trainees to advance to WTO1 positions after obtaining Grade 3 licenses. This lowered the minimum qualifications for trainees which resulted in trainees at wider range of abilities and higher turnover.

In 2022, the step raises were eliminated from the 369D positions as part of labor negotiations allowing management to bring in external candidates or promote internal candidates at full top-out pay. In addition, the 2022 MOU required that Public Works utilize the Maintenance Helper position to hire employees that could later apply for M1 positions.

Three wastewater treatment trainees were hired in 2022 with the ability to promote to WTO1 after obtaining a Grade 3 license. One trainee obtained a Grade 4 license and exited the program. By the end of 2022, there were 10 trainee positions, 6 of which were filled and 4 were vacant. Plans were developed to establish a continuous registry for trainees to streamline the interview and selection process.

At the beginning of 2023, there were 10 trainee positions, six of which were filled and 4 were vacant. During the year, one of the trainees obtained his Grade 3 license and exited the program. Many trainees were hired throughout the year to fill both existing vacancies and new vacancies created from converted wastewater operator positions. However, several trainees exited the program without obtaining a license to go to other City positions.

Table 36 shows the yearly number of budgeted positions for each plant over the years.

Covid 19 Testing in Wastewater

The treatment plants submitted raw wastewater samples to two different labs throughout the year as part of a national initiative to correlate Covid19 concentration with number of human cases at no cost to the City. In 2023, the Biobot program transitioned to the State of Tennnessee for both treatment plants. The Stiles Facility began participating in the Wastewater SCAN program beginning about July 2023 receiving a monthly stipend for its efforts. The testing included SARS, RSV, Influenza, Monkey Pox, and Hepatitis among others.

FIGURES

TABLES